

Symbolic Regression Can Perform on Par with Tree-based Ensembles in Regression Problems

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Abstract. In machine learning, the state of the art for regression problems is currently represented by black-box ensembles of decision trees, such as Random Forest and eXtreme Gradient Boosting. Symbolic Regression, an Evolutionary Computation technique, can tackle regression problems and deliver human-readable models in the form of equations. However, the consensus in the field is that Symbolic Regression is generally less effective than tree-based approaches. Providing a fair comparison of these techniques is not straightforward, as they employ extremely different paradigms. In this work, we propose an empirical evaluation of tree-based ensembles and Symbolic Regression, employing state-of-the-art software libraries compared on a curated regression benchmark suite, using rigorous statistical tests to assess differences in performance. The results reveal that Symbolic Regression with commonly used default hyperparameters is indeed generally less performant than tree-based ensembles. Nevertheless, we show that when the final model selection method is changed and more computational resources are allocated to all techniques, Symbolic Regression becomes more competitive, often performing on par with the state of the art. Finally, we verify the interpretability of the models discovered by Symbolic Regression for selected problems: An expert analysis of the equations discovered during the 10-fold cross-validation uncovers common structural elements, eventually delivering a global form.

Keywords: Benchmarking · Extreme Gradient Boosting · Random Forest · Regression · Symbolic Regression

1 Introduction

Given a dataset $\mathcal{D} = \{(X_i, y_i)\}_{i=0}^s$, where $X_i \in \mathbb{R}^n$ is a vector of n input features and $y_i \in \mathbb{R}$ is the ground truth value for each of the s samples, regression algorithms aim to find a function $\hat{y} = f(X)$ such that $y_i \approx \hat{y}_i, \forall i$. A considerable variety of approaches have been developed to tackle regression problems, from linear regression [20] to Support Vector Machines [7], from Decision Trees (DTs) [4] to ensembles of DTs such as Random Forest (RF) [3] and eXtreme Gradient Boosting (XGBoost) [6].

While most of the high-performing approaches originate from the field of Machine Learning (ML), at least one regression algorithm emerged from the domain of Evolutionary Computation (EC) and Genetic Programming (GP): Symbolic Regression (SR) [15], able to stochastically explore the space of free-form equations to discover interpretable models. Other techniques, such as Deep Learning, have also been employed to perform equation discovery [25], but in this work we will consider just the GP-based version, as it currently represents the dominant paradigm. Modern GP approaches to SR [5,8,23] have been successful in finding interpretable models for a wide range of applications, from industrial fouling [14], to bacterial growth [24], to associations between biomarkers and diseases [22].

Despite a remarkable history of favorable results, the general consensus among industry and practitioners for tabular regression problems is that SR is not among the state of the art in the domain, currently considered to be represented by tree-based ensembles such as RF and XGBoost [10]. For a long time, the main obstacle to a wider adoption of SR was the absence of software libraries usable out of the box, such as `scikit-learn` [21] for ML algorithms; but in recent years, several highly curated software solutions emerged, for example `Operon` [5] and `PySR` [8]. Another obstacle to the popularity of SR is the higher computational cost, order of magnitudes higher than RF or XGBoost; SR libraries often set their default hyperparameters so that SR terminates in a few seconds, a time comparable to other approaches, but this can go against the final quality of the results. Finally, modern SR software employs a complexity/fitting Pareto approach, but often the user just wants one single solution; and selecting a candidate model on the front is not easy, as heuristics might deliver under-performing results.

The motivation of this work is to design and carry out a proper, thorough experimental comparison of SR against modern regression algorithms, to verify whether SR can perform on par with the state of the art. To achieve our objective, we explore the following research questions:

- RQ1** Is it possible to design an experimental comparison which is as fair as possible for both SR and tree-based models, allowing them to efficiently exploit the same amount of computational resources?
- RQ2** Given enough computational resources, can SR be competitive with tree-based models for regression problems?
- RQ3** For SR, the final model selection on the complexity/fitting front is pivotal for the quality of the results. Is there a significant difference between using heuristics (as in modern SR libraries) and exploiting a validation set?

Tackling **RQ1** is not straightforward, as differences in algorithms (iterative versus parallel), implementation, and programming language can have significant impacts. As a proposed solution, we design an experimental evaluation around the OpenML-CTR23 benchmark suite [9], using the wall-clock time employed by SR to perform hyperparameter tuning for the tree-based models, with the hyperparameter optimization library `Optuna` [1].

The answer to **RQ2** is positive: During the experimental evaluation, the state-of-the-art SR library `PySR` [8] is often able to perform on par with RF

and XGBoost, even when hyperparameter tuning for the other algorithms is involved. However, obtaining this result comes with a considerable increase in computational time, allowing PySR to generate equations of large size (thus losing interpretability), and running it for a large amount of iterations.

Finally, comparison of SR models selected using PySR internal heuristic and a validation set carved out from the training data (thus reducing the amount of samples available for pure training) shows the second approach to be vastly superior, answering **RQ3** and providing a useful recommendation for practitioners interested in using SR for real-world problems.

As one of the main stated advantages of SR over other ML approaches is the interpretability of its results, in the last part of the paper we present the equations discovered by SR for selected datasets in the benchmark suite, showing how it is possible to eventually find a common form through an expert analysis, and obtain compact, interpretable, highly predictive models.

2 Background

2.1 Symbolic regression

Symbolic Regression (SR) is a technique designed for regression problems, from the field of Evolutionary Computation (EC), and Genetic Programming (GP) [15] in particular. SR discovers analytical mathematical expressions that describe underlying data patterns. Unlike traditional linear or quadratic regression, which fit parameters to a predefined functional form, SR searches the combinatorial space of mathematical operators, constants, and variables, to identify the model structure itself. Modern approaches to SR introduced improvements in the operators and the selection, for example preserving a set of equations each representing a different compromise on the complexity/error front [23]. After the initial release of commercial SR software such as Eureqa® Formulize in the 2010s, in recent years the SR community released highly optimized open-source software libraries, such as `Operon` [5] and `PySR` [8].

2.2 Ensembles of decision trees

Tree-based models are currently considered the state of the art for tabular regression [10]. Such approaches consistently outperform Deep Learning and other statistical models on standard tabular benchmarks. While recent Transformer-based approaches (such as TabPFN) have reported competitive or superior performance [12], we decided to not include them in this study, since (i) the benchmark suite employed here is likely to have been part of its training set, and (ii) a pre-trained model would not benefit from the hyperparameter optimization strategy described in the following.

A Decision Tree (DT) addresses regression by recursively partitioning the feature space into axis-aligned regions through a sequence of feature-threshold tests. Each terminal leaf is associated with a prediction value. Splits are selected

to minimize a loss or impurity criterion, but unconstrained trees can easily overfit unless regularized through depth limits, minimum leaf size, or pruning. To improve performance, modern approaches rely on ensembles of DTs.

Random Forest (RF) [3] constructs multiple trees using bootstrap sampling of the training data and random subsets of features at each split. Predictions are then aggregated across trees, typically by averaging for regression or majority voting for classification, which reduces variance and improves generalization. eXtreme Gradient Boosting (XGBoost) [6] builds trees sequentially, with each new tree trained to correct the residual errors of the current ensemble by optimizing a differentiable loss function. It combines regularization with efficient optimization strategies to improve predictive accuracy while controlling overfitting.

RF and XGBoost often achieve remarkable predictive performance by combining a large number of weak learners. However, this predictive strength comes at the cost of interpretability: unlike a single symbolic expression, such ensembles cannot be directly analyzed, algebraically manipulated, or checked for dimensional consistency. By contrast, SR provides interpretability through explicit analytical expressions, making it attractive when model transparency is required.

3 Proposed approach

3.1 The OpenML-CTR23 benchmark suite

While the SR community has also developed benchmarks suites for regression problems [16,19], in this study we decided to use the OpenML-CTR23 benchmark suite [9], which is commonly used in the field of Machine Learning (ML). OpenML-CTR23 includes 35 curated datasets, handpicked from the larger OpenML repository¹ [2], with a number of samples ranging from 517 to 72,000, and a number of features between 5 and 116. Benchmarks include mostly real-valued features, with a number of categorical features ranging from 0 to 17. Each dataset in the benchmark is connected to a reference, and folds for a k -fold cross-validation (typically with $k = 10$) are pre-computed to improve fairness in comparison. The datasets in the suite are detailed in Table 1.

3.2 Evaluating performance in regression

Regression performance can be assessed by evaluating metrics such as mean squared error and R^2 on the test sets of each fold. Assuming large differences in behavior between regressors and small standard deviation, comparing the mean R^2 on test could suffice; but in several cases, performances in state-of-the-art algorithms will be close. It is possible to establish whether a difference is statistically significant by employing hypothesis tests. In this setting, an algorithm’s performance on each fold is treated as a sample drawn from an underlying random distribution, and the test evaluates whether two sets of samples are likely to originate from the same distribution. The decision is then

¹ OpenML, <https://www.openml.org/>

Task ID	Dataset name	Samples	Features	Missing values	Categorical features
361234	abalone	4,177	8	0	1
361235	airfoil_self_noise	1,503	5	0	0
361236	australian_credit_evaluation	2,043	7	0	2
361237	concrete_compressive_strength	1,030	8	0	0
361241	physicochemical_protein	45,700	9	0	0
361242	superconductivity	21,263	81	0	0
361243	geographical_origin_of_music	1,059	116	0	0
361244	solar_flare	1,066	10	0	8
361247	naval_propulsion_plant	11,934	14	0	0
361249	white_wine	4,898	11	0	0
361250	red_wine	1,599	11	0	0
361251	grid_stability	10,000	12	0	0
361252	video_transcoding	68,784	18	0	2
361253	wave_energy	72,000	48	0	0
361254	sonar	48,933	21	0	0
361255	california_housing	20,640	8	0	0
361256	cpu_activity	8,192	21	0	0
361257	diamonds	53,940	9	0	3
361258	kidsum	8,192	8	0	0
361259	pumadyn32nh	8,192	32	0	0
361260	miss_housing	13,932	15	0	0
361261	cps84wages	28,155	6	0	4
361264	socomb	1,156	5	0	4
361266	kings_county	21,613	21	0	4
361267	brasilian_houses	10,002	9	0	4
361268	fps_benchmark	24,624	43 (39)	69,696	13
361269	health_insurance	22,272	11	0	7
361272	fifa	19,178	28	0	1
361616	Moneyball	1,232	14 (10)	3,600	4
361617	energy_efficiency	768	8	0	0
361618	forest_fire	517	12	0	2
361619	student_performance_por	649	30	0	17
361621	QSAR_fish_toxicity	908	6	0	0
361622	cars	804	17	0	0
361623	space_ga	3,107	6	0	0

Table 1. Regression tasks and related datasets included in the OpenML-CTR23 benchmark suite. Two datasets (**fps_benchmark** and **Moneyball**) have features with missing values, which will be removed during preprocessing.

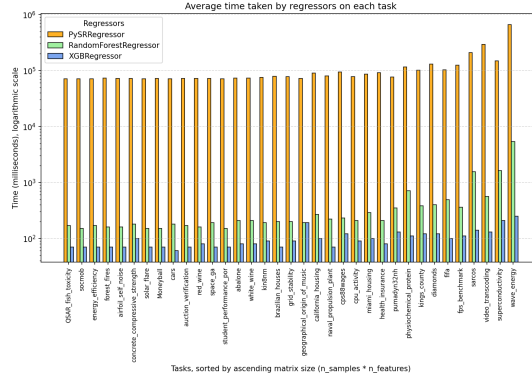


Fig. 1. Wall-clock time (milliseconds) taken by the considered algorithms on the different datasets. Datasets are sorted by size of the feature matrix (number of samples times number of features).

based on the p-value, denoted by p , that represents the probability of obtaining an observed difference at least as large as the measured one purely by chance, assuming that the null hypothesis is true, i.e., assuming that no real difference exists between the compared algorithms. A user-defined significance threshold, typically $\alpha = 0.05$, is then used: if $p < \alpha$, the null hypothesis is rejected and the difference is considered statistically significant.

Choosing the correct statistical test to apply is not straightforward, as a classic paired t -test for example assumes that the samples are independent; however, this is not true when performing a k -fold cross-validation, as most of the training set will overlap between different folds. To tackle this issue, we select a test specifically designed for such cases, the Nadeau-Bengio’s Corrected Resampled t -test [18], which introduces a correction factor to the variance calculation to account for the overlapping training data.

While Nadeau-Bengio works perfectly for evaluating the difference in performance between two regressors, a comparison between three or more regressors introduces additional subtleties. Selecting an arbitrary threshold α to establish significance means accepting a corresponding α chance of a false positive (Type I error) for a single test, for example $\alpha = 0.05$ would correspond to a $P(E) = 0.05$ chance of error; but since comparing multiple regressors implies performing several pairwise tests, with m regressors the true probability of obtaining at least one error becomes $P(E) = 1 - (1 - \alpha)^m$, going up to $P(E) = 0.143$ for $m = 3$. To solve this issue, we employ a Holm-Bonferroni correction [13].

3.3 Obstacles to a fair comparison

Ensuring fairness during the comparison of the different regression algorithms is quite challenging, for two reasons: (i) the inner working of each algorithm; (ii) the specific implementation.

SR and XGBoost are both iterative algorithms, while RF generates all its trees in parallel, at the same time. Furthermore, SR requires a full evaluation of each candidate solution generated, while RF and XGBoost both employ a greedy strategy at the level of each single leaf, relying upon faster assessments. Given these premises, it is easy to see how SR would benefit the most from a larger amount of available computational resources.

Since we are interested in comparing the state of the art, we rely upon established libraries for each of the considered algorithms. While the most proper way of ensuring fairness would be to re-implement all algorithms in the same programming language, the effort required would be immense, and the risk of missing important details introduced in the most up-to-date libraries would be extremely high. Using existing software, however, comes with several disadvantages.

The RF implementation employed in the study is `RandomForestRegressor` from the `scikit-learn` [21] Python module; for XGBoost, we use the `xgboost` Python module, which, however, relies on a compiled C++ core called with Python bindings; and the library chosen for SR is `PySR` [8], again with a Python interface but a core written in Julia. The speed of the scripts of an interpreted language like Python is clearly orders of magnitude slower than the compiled code of C++ or Julia: This creates issues when the aim is to fairly compare the algorithms.

To tackle these issues, we propose the following solutions. Given the differences in inner working and implementation of the algorithms involved in the comparison, we resort to measuring the wall-clock time taken by each approach on the same hardware, for each specific task. The time employed by the slowest algorithm (predictably SR) will then be used as a reference. Since SR can likely profit the most from the availability of more computational resources, the same amount of time used by SR will be devoted to performing hyperparameter optimization for RF and XGBoost. Hyperparameter optimization or hyperparameter tuning is the process of finding the values of the hyperparameters which deliver the best performance, typically assessed in a cross-validation or with a validation set.

4 Experimental evaluation

Following the ideas developed in the previous section, we perform two separate sets of experiments on OpenML-CTR23. In the first, all algorithms employ default hyperparameters of the respective libraries. In the second one, `PySR` uses enhanced hyperparameters, leading to a much higher computational cost and consequent longer wall-clock time; all other algorithms exploit the same amount of time to perform hyperparameter optimization with a state-of-the-art library. All experiments are executed on the same hardware, an Intel® Xeon® w9-3475X

with 36 cores and 128 GB of RAM. The code developed for the experimental evaluation, in Python, is available on a public GitHub repository².

4.1 Preprocessing

OpenML-CTR23 already has pre-computed splits for k -fold cross-validation, which is always with $k = 10$; for a few datasets a 10x10-fold cross-validation is defined, giving the possibility of performing the 10-fold cross-validation 10 times; but in all those cases, we just used the first 10-fold division. Feature columns with missing values have been dropped, which is only relevant for the two datasets **fps_benchmark** (going from 43 to 39 features) and **Moneyball** (going from 14 to 10 features). Values of categorical features have been converted to consecutive integer values, in order of appearance. When a validation set was used, for both PySR model selection and tree-based model hyperparameter tuning, the same 20% of the training set was randomly assigned to validation. No normalization was performed on the features, as both SR and tree-based approaches typically do not require values in a specific range to be effective.

4.2 Default hyperparameters

In a first experimental run, PySR with default hyperparameters (as of version 1.5.9) is tested on a 10-fold cross-validation on all datasets in OpenML-CTR23. As most SR libraries, PySR has a considerable number of hyperparameters; here we report only the values that will be changed in the following, which are also the most impactful for the quality of the final results. **binary_operators**=[+, -, *, /] describes all operators with two arguments; **unary_operators**=[], on the other hand, includes operators such as cos and log, but by default the list is empty. **maxsize**=30 is the maximum complexity of an equation, set as the sum of nodes in the tree. **niterations**=100 is the maximum number of generations given to the evolutionary algorithm. PySR employs a multi-population scheme, by default with **npopulations**=31 populations of **populationsize**=27 individuals each, for a total population size of 837 candidate solutions. The default heuristic that PySR uses to select the final model between non-dominated solutions on the error/complexity front is by first computing a score corresponding to the negated derivative of the log-error with respect to complexity; and then by selecting the candidate model with the highest score among expressions with an error better than at least 1.5 times the most accurate model [8].

RandomForestRegressor and **XGBRegressor** also employ default hyperparameters from their respective libraries, with the most impactful for both being the number of trees in the ensemble **n_estimators**=100. By default, **RandomForestRegressor** expands trees until all leaves are pure, or each leaf contains **min_samples_split**=2 samples or less. All features are considered for the splits, and the criterion used to optimize the trees is the mean squared error with respect to the training data. **XGBRegressor** has a higher number of

² <https://github.com/albertotonda/symbolic-regression-vs-ensembles>

hyperparameters, with the most relevant besides number of estimators being maximum tree depth `max_depth=6`, learning rate `eta=0.3`, and L2 regularization weight `lambda=1`.

Timing Figure 1 reports the average time spent by each algorithm on a single fold of the cross-validation process, per dataset. Datasets are sorted by size of their training matrix (number of samples multiplied by the number of features). XGBoost shows a consistent performance, with time per fold ranging from 0.06 s on the smallest datasets to 0.25 s on the largest. Random Forest, which is implemented in Python and not in a compiled language, but that on the other hand is not an iterative algorithm, takes between 0.15 s and 5.35 s per fold. As expected, the SR algorithm is by far the slowest, with times ranging from 70.83 s to 654.05 s. As each candidate solution for SR needs to be tested on the available training samples, PySR’s computational effort increases as the number of samples grows. It is important to remark that SR can in principle employ batching, evaluating each individual only on a small subset of the training data, but for the experiments in this work we decided to not use this option, as we are interested in assessing how the algorithm scales as the number of samples in the training set varies.

Performance As previously described in subsection 3.2, we employ the proper statistical tests to assess significant differences in performance. First, in order to explore **RQ3**, we are interested in comparing the two different methodologies for performing model selection in SR; we thus apply the pairwise Nadeau-Bengio test on the dataset, with results summarized in Table 2. Choosing a SR model using the validation set is always equal to or better than the heuristic, so in the following we will use only this modality.

To start assessing the initial situation related to **RQ2**, we compare the performance of the regressors using default hyperparameters. The complete results are reported in Table 3, with two columns reporting whether one of the algorithms is statistically better using two arbitrary thresholds of significance ($\alpha = 0.05$ and $\alpha = 0.01$). One first interesting global consideration is that some of the tasks appear difficult for all regressors. R^2 values can range in $(-\infty, 1.0]$, with values close to 1.0 corresponding to the best predictions: The most performing algorithm for tasks such as 361618 (`forest_fires`) or 361244 (`solar_flare`), for example, has an R^2 value which is negative or barely above 0.0, indicating an extremely poor performance. This leads us to believe that for those datasets it might not be possible to efficiently predict the information for the target using the information contained in the features. In other cases, such as tasks 361619 (`student_performance_por`) or 361264 (`socmob`), all regressors show a considerable variance, indicating that some of the folds might be harder to tackle.

For $p < 0.05$, as expected, XGBoost is overall the significantly best algorithm for most tasks (10/35), followed by Random Forest (7/35), with SR last (3/35); leaving out 25/35 datasets for which there is no significantly best regressor. We also performed a second statistical analysis on each task, verifying whether it

Task ID	Dataset name	PySR R2, mean and std		Significant difference	
		Heuristic	Validation	$p < 0.05$	$p < 0.01$
361234	abalone	0.5074 ± 0.0253	0.5652 ± 0.0229	Yes	Yes
361235	airfoil_self_noise	0.5217 ± 0.1020	0.6364 ± 0.0620	Yes	Yes
361236	auction_verification	0.8363 ± 0.0354	0.8773 ± 0.0274	Yes	Yes
361237	concrete_compressive_strength	0.7498 ± 0.0484	0.8092 ± 0.0271	Yes	Yes
361241	physiochemical_protein	0.1282 ± 0.0077	0.2890 ± 0.0091	Yes	Yes
361242	superconductivity	0.6065 ± 0.0330	0.7113 ± 0.0203	Yes	Yes
361243	geographical_origin_of_music	0.0878 ± 0.0521	0.0709 ± 0.0456	No	No
361244	solar_flare	0.0857 ± 0.0342	-1.5347 ± 4.8692	No	No
361247	naval_propulsion_plant	0.6511 ± 0.1264	0.6970 ± 0.1294	Yes	Yes
361249	white_wine	0.1858 ± 0.0387	0.2952 ± 0.0447	Yes	Yes
361250	red_wine	0.1650 ± 0.0926	0.3503 ± 0.0299	Yes	Yes
361251	grid_stability	0.6438 ± 0.0718	0.7000 ± 0.0456	Yes	No
361252	video_transcoding	0.7068 ± 0.0732	0.7734 ± 0.0376	Yes	Yes
361253	wave_energy	0.7957 ± 0.0474	0.7978 ± 0.0460	No	No
361254	sarcos	0.9108 ± 0.0049	0.9272 ± 0.0031	Yes	Yes
361255	california_housing	0.4937 ± 0.0990	0.6127 ± 0.0405	Yes	Yes
361256	cpu_activity	0.9637 ± 0.0098	0.9713 ± 0.0070	Yes	Yes
361257	diamonds	0.9400 ± 0.0521	0.7899 ± 0.4962	No	No
361258	kin8nm	0.3753 ± 0.1030	0.5241 ± 0.0327	Yes	Yes
361259	pumadyn32nh	0.4383 ± 0.2303	0.4451 ± 0.2350	Yes	No
361260	miami_housing	0.7056 ± 0.0474	0.7696 ± 0.0305	Yes	Yes
361261	cps85wages	0.0962 ± 0.0178	0.2724 ± 0.0403	Yes	Yes
361264	socmob	0.4331 ± 1.0089	0.7661 ± 0.0908	No	No
361266	kings_county	0.6649 ± 0.0513	0.7188 ± 0.0414	Yes	Yes
361267	brazilian_houses	0.4645 ± 0.3103	0.6718 ± 0.3171	No	No
361268	fps_benchmark	0.4811 ± 0.0938	0.6023 ± 0.0713	Yes	Yes
361269	health_insurance	0.2797 ± 0.0093	0.3659 ± 0.0137	Yes	Yes
361272	fifa	0.6794 ± 0.0308	0.7548 ± 0.0333	Yes	Yes
361616	Moneyball	0.9281 ± 0.0137	0.9453 ± 0.0044	Yes	No
361617	energy_efficiency	0.9481 ± 0.0355	0.9547 ± 0.0293	No	No
361618	forest_fires	-11.4194 ± 31.2189	-0.2514 ± 0.3384	No	No
361619	student_performance_por	0.0931 ± 0.0593	0.2208 ± 0.1170	Yes	No
361621	QSAR_fish_toxicity	0.4079 ± 0.0838	0.5559 ± 0.0843	Yes	Yes
361622	cars	0.8891 ± 0.0312	0.9136 ± 0.0308	Yes	Yes
361623	space_ga	0.4662 ± 0.0667	0.5484 ± 0.0347	No	No

Table 2. Comparison in performance of model selection approaches (heuristic choice versus using a validation set) for SR using PySR. If the difference is significant ($p < 0.05$), the model with the best R^2 score is highlighted in orange.

was possible to isolate one regressor as the significantly worst. For $p < 0.01$, the least performing regressor was SR (16/35), with RF under-performing on 2/35 datasets and XGBoost on 1/35; for 16/35 datasets, there is no clear worst performer. From these results, it is clear that with default hyperparameters SR is not only slower, but also less performing than the tree-based models. In the next set of experiments, we aim to verify whether it is possible to improve SR’s performance by giving the algorithm more time to explore a vaster search space of equations, while simultaneously allowing the tree-based models to improve their performance by tuning hyperparameters for the same amount of time.

4.3 Enhanced hyperparameters

In a second experimental evaluation, PySR is manually set with hyperparameter values leading to a larger search space and more computationally expensive runs: `binary_operators=[+, -, *, /]`, `unary_operators=[sin, cos, log, exp]`, `maxsize=300`, `populationsize=100` (for a total number of 3,100 individuals across all populations), `niterations=1000`. The average wall-clock time for each split in the same dataset (now much larger than in the previous experiments) is then used as a timeout for performing hyperparameter tuning on RF and XGBoost using Optuna [1]. We set the maximum allowed time for PySRRegressor to 24 minutes of wall-clock time per fold, to ensure that a full

Task ID	Dataset name	R2, mean and std			Significant best Significant worst			
		Symbolic Regression	Random Forest	XGBoost	$p < 0.05$	$p < 0.01$	$p < 0.05$	$p < 0.01$
361234	abalone	0.5652 ± 0.0229	0.5443 ± 0.0177	0.4919 ± 0.0282	SR	-	XGB	XGB
361235	airfoil_self_noise	0.6364 ± 0.0620	0.9292 ± 0.0151	0.9466 ± 0.0152	XGB	XGB	SR	SR
361236	auction_verification	0.8773 ± 0.0274	0.9928 ± 0.0015	0.9983 ± 0.0005	XGB	XGB	SR	SR
361237	concrete_compressive_strength	0.8092 ± 0.0271	0.8995 ± 0.0274	0.9232 ± 0.0240	XGB	XGB	SR	SR
361241	physiochemical_protein	0.2890 ± 0.0091	0.6624 ± 0.0057	0.6015 ± 0.0057	RF	RF	SR	SR
361242	superconductivity	0.7113 ± 0.0203	0.9217 ± 0.0074	0.9176 ± 0.0066	RF	RF	SR	SR
361243	geographical_origin_of_music	0.0709 ± 0.0456	0.2623 ± 0.0922	0.1841 ± 0.0935	RF	-	SR	-
361244	solar_flare	-1.5347 ± 4.8692	-0.0980 ± 0.1430	-0.2817 ± 0.2452	-	-	-	-
361247	naval_propulsion_plant	0.6970 ± 0.1294	0.9954 ± 0.0006	0.9926 ± 0.0006	RF	RF	SR	SR
361249	white_wine	0.2952 ± 0.0447	0.5097 ± 0.0387	0.4688 ± 0.0356	RF	RF	SR	SR
361250	red_wine	0.3503 ± 0.0299	0.4717 ± 0.0564	0.4308 ± 0.0514	RF	RF	SR	SR
361251	grid_stability	0.7000 ± 0.0456	0.8944 ± 0.0074	0.9294 ± 0.0071	XGB	XGB	SR	SR
361252	video_transcoding	0.7734 ± 0.0376	0.9809 ± 0.0014	0.9865 ± 0.0013	RF	RF	SR	SR
361253	wave_energy	0.7978 ± 0.0460	0.8328 ± 0.0036	0.9579 ± 0.0013	XGB	XGB	-	-
361254	sarcos	0.9272 ± 0.0031	0.9765 ± 0.0013	0.9759 ± 0.0011	-	-	SR	SR
361255	california_housing	0.6127 ± 0.0405	0.8165 ± 0.0139	0.8279 ± 0.0143	XGB	XGB	SR	SR
361256	cpu_activity	0.9713 ± 0.0070	0.9812 ± 0.0049	0.9815 ± 0.0067	-	-	SR	-
361257	diamonds	0.7899 ± 0.4962	0.9814 ± 0.0011	0.9812 ± 0.0011	-	-	-	-
361258	kin8nm	0.5241 ± 0.0327	0.7010 ± 0.0176	0.7666 ± 0.0161	XGB	XGB	SR	SR
361259	pumadyn32nh	0.4451 ± 0.2350	0.6356 ± 0.0190	0.5911 ± 0.0157	-	-	-	-
361260	miami_housing	0.7696 ± 0.0305	0.9059 ± 0.0149	0.9125 ± 0.0147	XGB	-	SR	SR
361261	cps88wages	0.2724 ± 0.0403	0.1720 ± 0.0363	0.2591 ± 0.0427	SR	-	RF	RF
361264	socmob	0.7661 ± 0.0908	0.7326 ± 0.2032	0.7815 ± 0.2623	-	-	-	-
361266	kings_county	0.7188 ± 0.0414	0.8681 ± 0.0492	0.8731 ± 0.0294	-	-	SR	SR
361267	brazilian_houses	0.6718 ± 0.3171	0.5053 ± 0.2843	0.3622 ± 0.6316	-	-	-	-
361268	fps_benchmark	0.6023 ± 0.0713	0.9958 ± 0.0005	0.9979 ± 0.0002	XGB	XGB	SR	SR
361269	health_insurance	0.3659 ± 0.0137	0.3155 ± 0.0204	0.3652 ± 0.0134	-	-	RF	RF
361272	fifa	0.7548 ± 0.0333	0.7687 ± 0.0298	0.7577 ± 0.0294	-	-	-	-
361616	Moneyball	0.9453 ± 0.0044	0.9262 ± 0.0082	0.9256 ± 0.0043	SR	SR	-	-
361617	energy_efficiency	0.9547 ± 0.0293	0.9975 ± 0.0011	0.9988 ± 0.0005	XGB	-	SR	-
361618	forest_fires	-0.2514 ± 0.3384	-4.6276 ± 8.4307	-11.4936 ± 16.6271	-	-	-	-
361619	student_performance_por	0.2208 ± 0.1170	0.2158 ± 0.1596	0.0479 ± 0.2114	-	-	-	-
361621	QSAR_fish_toxicity	0.5559 ± 0.0843	0.6011 ± 0.0669	0.5593 ± 0.0681	-	-	-	-
361622	cars	0.9136 ± 0.0308	0.9358 ± 0.0270	0.9333 ± 0.0199	-	-	-	-
361623	space_ga	0.5484 ± 0.0347	0.6487 ± 0.0618	0.6777 ± 0.0463	-	-	-	-

Table 3. Performance of selected regressors with default hyperparameters.

10-fold cross-validation using all regressors will be completed in 12 hours, and thus a full experimental run on OpenML-CTR23 will last a maximum of 18 days. The hyperparameter search space for the ensembles is detailed in Appendix C of the Supplementary Materials³.

Performance The results of the different regressors with enhanced hyperparameters are reported in Table 4. The difference between algorithms is reduced: For $p < 0.05$, XGBoost is now the best in 8 tasks (down from 10), RF in 6 (down from 7) and SR in 2 (down from 3); as for the least-performing algorithms, at $p < 0.01$, SR is now the worst for 14 tasks (down from 16), XGBoost is the worst in 1 (same as before), while interestingly RF is never the worst (down from 2).

Results reported in Table 5 show that the R^2 values generally improve significantly with enhanced hyperparameters, typically with larger gains for SR.

5 Discussion

5.1 LLM-assisted expert analysis of SR models

One of the stated advantages of SR over other black-box ML regression algorithms is the interpretability of the results. In order to evaluate this claim, we perform

³ Supplementary materials,

Task ID	Dataset name	R2, mean and std			Significant best Significant worst			
		Symbolic Regression	Random Forest	XGBoost	$p < 0.05$	$p < 0.01$	$p < 0.05$	$p < 0.01$
361234	abalone	0.5763 ± 0.0204	0.5615 ± 0.0192	0.5499 ± 0.0276	-	-	-	-
361235	airfoil_self_noise	0.8689 ± 0.0188	0.9324 ± 0.0153	0.9438 ± 0.0190	-	-	SR	SR
361236	auction_verification	0.9588 ± 0.0290	0.9928 ± 0.0015	0.9983 ± 0.0005	XGB	-	SR	-
361237	concrete_compressive_strength	0.8612 ± 0.0405	0.9046 ± 0.0287	0.9317 ± 0.0201	XGB	XGB	SR	SR
361241	physiochemical_protein	0.2813 ± 0.0142	0.6718 ± 0.0051	0.6771 ± 0.0118	-	-	SR	SR
361242	superconductivity	0.7112 ± 0.0203	0.9244 ± 0.0064	0.9234 ± 0.0081	-	-	SR	SR
361243	geographical_origin_of_music	0.0334 ± 0.1852	0.2778 ± 0.0530	0.2781 ± 0.0663	-	-	SR	-
361244	solar_flare	0.0676 ± 0.0820	0.1241 ± 0.0372	0.1184 ± 0.0324	-	-	-	-
361247	naval_propulsion_plant	0.5574 ± 0.0927	0.9957 ± 0.0006	0.9926 ± 0.0006	RF	RF	SR	SR
361249	white_wine	0.3298 ± 0.0441	0.5271 ± 0.0385	0.4748 ± 0.0400	RF	RF	SR	SR
361250	red_wine	0.3723 ± 0.0358	0.4779 ± 0.0517	0.4175 ± 0.0405	RF	-	SR	-
361251	grid_stability	0.8148 ± 0.0400	0.8970 ± 0.0068	0.9294 ± 0.0071	XGB	XGB	SR	SR
361252	video_transcoding	0.6751 ± 0.0579	0.9911 ± 0.0011	0.9921 ± 0.0019	-	-	SR	SR
361253	wave_energy	0.8374 ± 0.1784	0.8397 ± 0.0048	0.9687 ± 0.0124	-	-	-	-
361254	sarcos	0.9286 ± 0.0045	0.9796 ± 0.0013	0.9845 ± 0.0016	XGB	XGB	SR	SR
361255	california_housing	0.6292 ± 0.0637	0.8214 ± 0.0147	0.8351 ± 0.0115	XGB	-	SR	SR
361256	cpu_activity	0.9677 ± 0.0206	0.9818 ± 0.0049	0.9822 ± 0.0072	-	-	-	-
361257	diamonds	0.9565 ± 0.0199	0.9818 ± 0.0011	0.9811 ± 0.0010	RF	-	SR	-
361258	kin8nm	0.7440 ± 0.0435	0.7106 ± 0.0180	0.7666 ± 0.0161	-	-	-	-
361259	pumadyn32nh	0.6585 ± 0.0159	0.6474 ± 0.0176	0.5911 ± 0.0157	SR	SR	XGB	XGB
361260	miami_housing	0.8152 ± 0.0364	0.9103 ± 0.0137	0.9223 ± 0.0128	XGB	XGB	SR	SR
361261	cps88wages	0.2850 ± 0.0422	0.2911 ± 0.0465	0.2715 ± 0.0403	-	-	-	-
361264	socmob	0.7463 ± 0.3725	0.7347 ± 0.1798	0.8002 ± 0.1930	-	-	-	-
361266	kings_county	0.7642 ± 0.0301	0.8705 ± 0.0425	0.8847 ± 0.0353	-	-	SR	SR
361267	brazilian_houses	0.6937 ± 0.2696	0.3828 ± 0.3372	0.2067 ± 0.5606	-	-	-	-
361268	fps_benchmark	0.6586 ± 0.0613	0.9959 ± 0.0005	0.9990 ± 0.0003	XGB	XGB	SR	SR
361269	health_insurance	0.3818 ± 0.0112	0.4038 ± 0.0123	0.3738 ± 0.0195	RF	RF	-	-
361272	fifa	0.7593 ± 0.0301	0.7730 ± 0.0297	0.7665 ± 0.0248	-	-	-	-
361616	Moneyball	0.9433 ± 0.0069	0.9292 ± 0.0069	0.9329 ± 0.0062	SR	-	-	-
361617	energy_efficiency	0.9970 ± 0.0010	0.9976 ± 0.0010	0.9988 ± 0.0005	XGB	XGB	SR	SR
361618	forest_fires	-8.1797 ± 23.5817	-0.4277 ± 0.7631	-4.4231 ± 4.9086	-	-	-	-
361619	student_performance_por	0.1441 ± 0.1645	0.2861 ± 0.1282	0.2337 ± 0.2212	-	-	-	-
361621	QSAR_fish_toxicity	0.4196 ± 0.4089	0.6254 ± 0.0534	0.5933 ± 0.0569	-	-	-	-
361622	cars	0.9449 ± 0.0193	0.9406 ± 0.0182	0.9388 ± 0.0202	-	-	-	-
361623	space_ga	0.5853 ± 0.0714	0.6596 ± 0.0621	0.6777 ± 0.0463	-	-	SR	-

Table 4. Performance of selected regressors with enhanced hyperparameters.

an expert analysis of the equations found during the previous experiments. As SR was run in a 10-fold cross-validation, 10 different equations were obtained for each dataset, with potentially considerable differences in structure. Our objective, for each dataset considered, is to uncover a generic equation, extracting recurring elements found in the 10 SR equations. Examining each equation to find common structural patterns by hand can be a long process, so we decide to include a Large Language Model (LLM) in the process, to perform a first screening prior to human intervention.

More in detail, the proposed expert analysis includes the following steps: (i) L^AT_EX files with the extracted equations are uploaded to a commercial LLM, along with a prompt asking to extract the common parts and find a generic equation with undefined constants $c_1, \dots, c_k$; (ii) one of the authors reviews the text produced by the LLM, and eventually corrects the shape of the generic equation; (iii) the constants are optimized first using the whole dataset, to have a global equation, and then in a 10-fold cross-validation to assess overfitting. The LLM selected for the experiments is the free version of Google Gemini[®] 3.1, accessed between May 19 and June 6, 2026, using the “Pro” mode with the “Extended reasoning” option. The constants have been optimized with the Covariance-Matrix-Adaptation Evolution Strategy (CMA-ES) [11], with default hyperparameters except population size set to $\mu = 100$ to improve repeatability.

We choose to analyze only the datasets for which SR obtains a mean $R^2 > 0.6$. The full results are reported in Table 6, and here in the following we highlight a

Task ID	Dataset name	R2 PySRRegressor (Heuristic)		R2 PySRRegressor (Validation)		R2 RandomForestRegressor		R2 XGBRegressor	
		Default	Enhanced	Default	Enhanced	Default	Enhanced	Default	Enhanced
361234	abalone	0.5074 ± 0.0253	0.4975 ± 0.0365	0.5652 ± 0.0229	0.5763 ± 0.0204	0.5443 ± 0.0177	0.5615 ± 0.0192	0.4919 ± 0.0282	0.5499 ± 0.0276
361235	airfoil_self_noise	0.5217 ± 0.1020	0.8412 ± 0.0223	0.6364 ± 0.0620	0.8689 ± 0.0188	0.9292 ± 0.0151	0.9324 ± 0.0153	0.9466 ± 0.0152	0.9438 ± 0.0190
361236	auction_verification	0.8363 ± 0.0354	0.8576 ± 0.0167	0.8773 ± 0.0274	0.9588 ± 0.0290	0.9928 ± 0.0015	0.9928 ± 0.0015	0.9983 ± 0.0005	0.9983 ± 0.0005
361237	concrete_compressive_strength	0.7498 ± 0.0484	0.8473 ± 0.0417	0.8092 ± 0.0271	0.8612 ± 0.0405	0.8995 ± 0.0274	0.9046 ± 0.0287	0.9232 ± 0.0240	0.9317 ± 0.0201
361241	physiochemical_protein	0.1282 ± 0.0077	0.1282 ± 0.0076	0.2890 ± 0.0091	0.2813 ± 0.0142	0.6624 ± 0.0057	0.6718 ± 0.0051	0.6015 ± 0.0057	0.6771 ± 0.0118
361242	superconductivity	0.6065 ± 0.0330	0.6119 ± 0.0409	0.7113 ± 0.0203	0.7112 ± 0.0203	0.9217 ± 0.0074	0.9244 ± 0.0064	0.9176 ± 0.0066	0.9234 ± 0.0081
361243	geographical_origin_of_music	0.0878 ± 0.0521	-0.0274 ± 0.4768	0.0709 ± 0.0456	0.0334 ± 0.1852	0.2623 ± 0.0922	0.2778 ± 0.0530	0.1841 ± 0.0935	0.2781 ± 0.0663
361244	solar_flare	0.0857 ± 0.0342	0.0696 ± 0.0589	-1.5347 ± 4.8692	0.0670 ± 0.0820	-0.0980 ± 0.1430	0.1241 ± 0.0372	-0.2817 ± 0.2452	0.1184 ± 0.0324
361247	naval_propulsion_plant	0.6511 ± 0.1264	0.4132 ± 0.1367	0.6970 ± 0.1294	0.5574 ± 0.0927	0.9954 ± 0.0006	0.9957 ± 0.0006	0.9926 ± 0.0006	0.9926 ± 0.0006
361249	white_wine	0.1858 ± 0.0387	0.1650 ± 0.0360	0.2952 ± 0.0447	0.3298 ± 0.0441	0.5097 ± 0.0387	0.5271 ± 0.0385	0.4688 ± 0.0356	0.4748 ± 0.0400
361250	red_wine	0.1650 ± 0.0926	0.2407 ± 0.0903	0.3503 ± 0.0299	0.3723 ± 0.0358	0.4717 ± 0.0564	0.4779 ± 0.0517	0.4308 ± 0.0514	0.4175 ± 0.0405
361251	grid_stability	0.6438 ± 0.0718	0.7647 ± 0.0491	0.7000 ± 0.0456	0.8148 ± 0.0400	0.8944 ± 0.0071	0.8970 ± 0.0068	0.9294 ± 0.0071	0.9294 ± 0.0071
361252	video_transcoding	0.7068 ± 0.0732	0.5974 ± 0.0683	0.7734 ± 0.0376	0.6751 ± 0.0579	0.9899 ± 0.0014	0.9911 ± 0.0011	0.9865 ± 0.0013	0.9621 ± 0.0019
361253	wave_energy	0.7957 ± 0.0474	0.8090 ± 0.2395	0.7978 ± 0.0460	0.8374 ± 0.1784	0.8328 ± 0.0036	0.8397 ± 0.0048	0.9579 ± 0.0013	0.9687 ± 0.0124
361254	sarcos	0.9108 ± 0.0049	0.9100 ± 0.0099	0.9272 ± 0.0031	0.9286 ± 0.0045	0.9765 ± 0.0013	0.9796 ± 0.0013	0.9759 ± 0.0011	0.9845 ± 0.0016
361255	california_housing	0.4937 ± 0.0990	0.5461 ± 0.0935	0.6127 ± 0.0405	0.6292 ± 0.0637	0.8165 ± 0.0139	0.8214 ± 0.0147	0.8279 ± 0.0143	0.8351 ± 0.0115
361256	cpu_activity	0.9637 ± 0.0008	0.9647 ± 0.0089	0.9713 ± 0.0070	0.9677 ± 0.0296	0.9812 ± 0.0049	0.9818 ± 0.0049	0.9815 ± 0.0067	0.9822 ± 0.0072
361257	diamonds	0.9400 ± 0.0521	0.9457 ± 0.0283	0.7899 ± 0.4962	0.9565 ± 0.0199	0.9814 ± 0.0011	0.9818 ± 0.0011	0.9812 ± 0.0011	0.9811 ± 0.0010
361258	kin8nm	0.3753 ± 0.1030	0.6694 ± 0.0642	0.5241 ± 0.0327	0.7440 ± 0.0435	0.7010 ± 0.0176	0.7106 ± 0.0180	0.7666 ± 0.0161	0.7666 ± 0.0161
361259	pumadyn32nh	0.4383 ± 0.2303	0.6610 ± 0.0156	0.4451 ± 0.2350	0.6585 ± 0.0159	0.6356 ± 0.0190	0.6474 ± 0.0176	0.5911 ± 0.0157	0.5911 ± 0.0157
361260	miami_housing	0.7056 ± 0.0474	0.7371 ± 0.0530	0.7096 ± 0.0305	0.8152 ± 0.0364	0.9059 ± 0.0149	0.9103 ± 0.0137	0.9125 ± 0.0147	0.9223 ± 0.0128
361261	cpu8stages	0.0962 ± 0.0178	0.1397 ± 0.0630	0.2724 ± 0.0403	0.2850 ± 0.0422	0.1720 ± 0.0363	0.2911 ± 0.0465	0.2591 ± 0.0427	0.2715 ± 0.0403
361264	socomb	0.4331 ± 1.0089	0.7125 ± 0.3993	0.7661 ± 0.0908	0.7463 ± 0.3725	0.7326 ± 0.2032	0.7347 ± 0.1798	0.7815 ± 0.2623	0.8002 ± 0.1930
361266	kinge_county	0.6649 ± 0.0513	0.6908 ± 0.0359	0.7188 ± 0.0414	0.7642 ± 0.0301	0.8681 ± 0.0492	0.8705 ± 0.0425	0.8731 ± 0.0294	0.8847 ± 0.0333
361267	brazilian_houses	0.4645 ± 0.3103	0.6755 ± 0.2436	0.6718 ± 0.3171	0.6937 ± 0.2696	0.5053 ± 0.2843	0.3828 ± 0.3372	0.3622 ± 0.6316	0.2067 ± 0.5606
361268	fpu_benchmark	0.4811 ± 0.0938	0.5842 ± 0.0962	0.6023 ± 0.0713	0.6586 ± 0.0613	0.9958 ± 0.0005	0.9959 ± 0.0005	0.9979 ± 0.0002	0.9990 ± 0.0003
361269	health_insurance	0.2797 ± 0.0093	0.2586 ± 0.0111	0.3659 ± 0.0137	0.3818 ± 0.0112	0.3155 ± 0.0204	0.4038 ± 0.0123	0.3652 ± 0.0134	0.3738 ± 0.0195
361272	fifa	0.6794 ± 0.0308	0.7262 ± 0.0392	0.7548 ± 0.0333	0.7593 ± 0.0301	0.7687 ± 0.0298	0.7730 ± 0.0297	0.7577 ± 0.0294	0.7665 ± 0.0248
361616	Moneyball	0.9281 ± 0.0137	0.9383 ± 0.0091	0.9453 ± 0.0044	0.9433 ± 0.0069	0.9262 ± 0.0082	0.9292 ± 0.0069	0.9256 ± 0.0043	0.9329 ± 0.0062
361617	energy_efficiency	0.9481 ± 0.0355	0.9967 ± 0.0007	0.9547 ± 0.0293	0.9970 ± 0.0010	0.9975 ± 0.0011	0.9976 ± 0.0010	0.9988 ± 0.0005	0.9988 ± 0.0005
361618	forest_fires	-11.4194 ± 31.2189	-47.2201 ± 120.8610	-25.514 ± 0.3284	-8.1797 ± 23.5817	-4.6276 ± 8.4307	-0.4277 ± 0.7631	-11.4936 ± 16.6271	-4.4231 ± 4.9086
361619	student_performance_por	0.0931 ± 0.0593	0.0327 ± 0.3776	0.2208 ± 0.1170	0.1441 ± 0.1645	0.2158 ± 0.1596	0.2861 ± 0.1282	0.0479 ± 0.2114	0.2337 ± 0.2212
361621	QSAR_fish_toxicity	0.4079 ± 0.0838	0.3501 ± 0.4786	0.5559 ± 0.0843	0.4196 ± 0.4089	0.6011 ± 0.0669	0.6254 ± 0.0534	0.5593 ± 0.0681	0.5933 ± 0.0569
361622	cars	0.8891 ± 0.0312	0.9376 ± 0.0239	0.9136 ± 0.0308	0.9449 ± 0.0193	0.9358 ± 0.0270	0.9406 ± 0.0182	0.9333 ± 0.0199	0.9388 ± 0.0202
361623	space_ga	0.4662 ± 0.0667	0.5119 ± 0.0715	0.5484 ± 0.0347	0.5853 ± 0.0714	0.6487 ± 0.0618	0.6596 ± 0.0621	0.6777 ± 0.0463	0.6777 ± 0.0463

Table 5. Regressors’ performance, with default and enhanced hyperparameters. Highlighted cells are significantly better using the Nadeau-Bengio test for $p < 0.05$.

few interesting cases. In task 361259 (pumadyn32nh), SR performs significantly better than the ensembles. The equation resulting from the analysis is quite compact and elegant:

$$\hat{y} = -1.077 \cdot 10^{-3} x_{15} \cos(x_4), \quad R^2 = 0.6615 \quad (1)$$

The information on pumadyn32nh from the OpenML website⁴ states: “A realistic simulation of the dynamics of a Puma 560 robot arm. The task [...] is to predict the angular acceleration of one of the robot arm’s links. The task is nonlinear and has high noise. [...]”. It thus makes sense that SR, with cos and sin operators, is able to find a good correlation, while tree-based ensembles might struggle.

The equation found for task 361272 (fifa) uses a single feature, x_4 :

$$\hat{y} = 0.17(x_4 - 56.201)^4 + 1823.898, \quad R^2 = 0.7568 \quad (2)$$

From the the dataset description on the OpenML website⁵, we learn that the objective of the task is to predict the wage of soccer players in the videogame FIFA, given several in-game features, and that x_4 is called overall, with no more information. However, searching online reveals that overall is the overall best relative rank of the player, providing an intuitive relationship between rank and wages, where small increases in rank can have substantial impact on the player’s salary, with respect to a baseline (the large constant term 1823.898).

On the two tasks 361249 (white_wine) and 361250 (red_wine) SR performs significantly worse than the other approaches with both default and enhanced hyperparameters: Inspecting the two datasets reveals them to be quite unusual

⁴ <https://openml.org/search?type=data&status=active&id=44981>

⁵ <https://openml.org/search?type=data&id=45012>

among regression problems. Both present quantitative, real-valued features about wine (such as pH, density, content of sulfur dioxide, etc.), with the target $y \in \mathbb{N}^+$ being an assessment of quality performed by human experts. This quality assessment value ranges from 3 to 9 in `white_wine` and from 3 to 8 in `red_wine`. It is interesting to notice that the same tasks could be easily reframed as classification problems; and in fact, the ML domain defines problems of this kind as *ordinal classification* or *ordinal regression* [17], because the relative values of the integer labels have a meaning. RF and XGBoost employ similar base predictors, DTs, which ultimately predict constant values after partitioning the feature space, a characteristic which might make the tree-based ensembles more suited for these kinds of problems. Another consideration is that SR always employs MSE as its fitness function, while for ordinal regression other functions could be more suited.

Nevertheless, the highlighted differences show how different regressors might be inherently more suited for different tasks, reinforcing the conclusions of the *no free lunch* theorem for ML [26]. A more detailed description of the analysis process for each dataset is presented in Appendix E.

5.2 Limitations

The choice of employing SR with a wall-clock timeout (24 minutes in the experiments of Section 4) can have undesired effects: For example, when using enhanced hyperparameters, SR has a vaster search space to explore, a larger population, and more complex equations to evaluate, which naturally leads to a longer wall-clock time spent on each generation. In some cases, especially for datasets with a considerable number of samples, the timeout might prevent SR from efficiently exploring the search space, stopping the algorithm after just a few generations. Looking back at Table 4, this is what likely happens for tasks 361247 and 361252, the only two cases for which SR performs significantly *worse* with enhanced hyperparameters than with default hyperparameters. As a sanity check, we re-run the two experiments, this time employing a 5-hour timeout for SR, and we obtain a significant improvement: SR’s performance on 361247 goes from $R2 = 0.5574 \pm 0.0927$ to $R2 = 0.7108 \pm 0.1223$ and on 361252 goes from $R2 = 0.6751 \pm 0.0579$ to $R2 = 0.8491 \pm 0.0288$.

6 Conclusions

This work presented a thorough empirical evaluation of SR against state-of-the-art tree-based models, with the objectives of **(RQ1)** designing an experimental setup as fair as possible for all regressors, **(RQ2)** assessing whether SR can perform on par with the state of the art if given enough computational resources, and **(RQ3)** evaluating a heuristic choice against using a validation set for the final choice of model for SR.

The results on a benchmark suite designed for ML regression techniques show that SR can be competitive with RF and XGBoost: Using default hyperparameters commonly found in software libraries, it often performs significantly worse, but

Task ID	Equation	Equation R2	Highest R2 (Algorithm)
361236	$\hat{y} = 92.431x_4 + \left(25.733x_3 - 8.269x_5 + 51.408 + \frac{36.008}{1.286 - x_5}\right) \cdot \left(-72.759x_0^2 + 220.591x_0 - 1.82x_4 + 131.606 - \frac{414.259}{x_0 + 274.477}\right) - 6061.062$	0.8901 ± 0.0223	0.9983 ± 0.0005 (XGB)
361237	$\hat{y} = 13.342 - \frac{739.372}{x_7 + 17.038} + \frac{627.204x_0 + 477.672x_1 + 379.5x_2}{27.052x_3 + 801.264}$	0.8154 ± 0.0272	0.9317 ± 0.0201 (XGB)
361242	$\hat{y} = 4.94x_{25}x_{77} - 21.151x_{25} + 18.978x_{66} - 3.351x_{77} + \frac{1010.188x_{27} + x_{70}(19.412x_{27} + 92.837x_9 - 2207.771) - 4663.478x_{26} + 54.219x_{44} + (123.637x_{72} - 216.296)(59.175x_{73} - 42.878) + 5527.946}{54.219x_{44} + (123.637x_{72} - 216.296)(59.175x_{73} - 42.878) + 5527.946} + 11.917$	0.7217 ± 0.0143	0.9217 ± 0.0074 (XGB)
361247	$\hat{y} = 0.028x_{10} + 1.21x_{11} - 0.003x_8 + \frac{3304.762x_{11} - 3446.023}{x_8(3.953x_9 - 9.853) + 3678.592} + 1.133$	0.8020 ± 0.0035	0.9957 ± 0.0006 (RF)
361251	$\hat{y} = -0.0997 \sin(1.124 \sin(0.0750x_1x_9 + 0.06619x_{10}x_3 + 0.3753x_8 + 3.2948 + \frac{2.3662e^{-0.6165x_{11}} \log(1.0000 - 5.6576 \cdot 10^{-9}x_2)}{-1.9512 \cdot 10^{-8}x_0 - 2.0206 \cdot 10^{-7}}))$	0.6499 ± 0.0255	0.9294 ± 0.0071 (XGB)
361252	$\hat{y} = \frac{9.825 \cdot 10^{-4}x_{13}^3x_{16} - 0.2125}{x_{13} - 0.9828 + \frac{4.8670x_{16}}{x_{14}}} + 0.9912$	0.7823 ± 0.0112	0.9921 ± 0.0019 (XGB)
361253	$\hat{y} = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - 105.301$	1.0000 ± 0.0000	0.9687 ± 0.0124 (XGB)
361254	$\hat{y} = 37.795x_0 - x_{10} + 1.149x_{14} - 2.975x_7x_8 + 20.163 + \frac{2.273x_{14} + 1.098x_{17}}{x_9}$	0.9312 ± 0.0026	0.9759 ± 0.0011 (XGB)
361256	$\hat{y} = -0.058x_{15} - 0.03x_{17} + \frac{0.633}{1.354 \cdot 10^{-7}x_2 + \frac{1}{x_{29}} + 0.006}$	0.9708 ± 0.0068	0.9815 ± 0.0067 (XGB)
361257	$\hat{y} = (21.035x_0 - 3.686) \cdot (x_6 - 1.788) \cdot (0.605x_3(-4.107x_2 + 3.456x_6 + 4.696) + 40.006)$	0.9618 ± 0.0023	0.9812 ± 0.0011 (XGB)
361259	$\hat{y} = -1.077 \cdot 10^{-3}x_{15} \cos(x_4)$	0.6612 ± 0.0157	0.6585 ± 0.0159 (SR)
361264	$\hat{y} = 6.325x_3 + 0.936x_4 - \frac{0.24x_6}{x_1 - 3.701}$	0.7597 ± 0.0859	0.8002 ± 0.1930 (XGB)
361266	$\hat{y} = 31.867x_{11} + 74.53x_{12} + (20.352x_6 + 44.071x_8 - 226.401) \cdot (42.249x_{14} + x_2 + 2397.756x_5 - 2121.226) + 3828.561 \sin(21.852x_{14} + 174.102) + 1669.555$	0.6434 ± 0.0431	0.8847 ± 0.0353 (XGB)
361267	$\hat{y} = 393.222x_0 + 1399.873x_3 + 571.234x_4 + x_8 - 884.748$	0.7004 ± 0.2379	0.6718 ± 0.3171 (SR)
361268	$\hat{y} = 3.431x_1 + 2.782x_{20} + 0.873x_{29} + 0.333x_{30}x_{38} + 1.946x_{30} - 35.873x_{38}^2 + 39.939 \cos(25.355 \sin(6.623x_{36} - 3.141)) + 444.305 + \frac{371.582}{x_{36} + 4.951} - \frac{20.649}{x_{36} - 0.596} - \frac{380.464}{x_{38}}$	0.7066 ± 0.0099	0.9990 ± 0.0003 (XGB)
361272	$\hat{y} = 0.17(x_4 - 56.201)^4 + 1823.898$	0.7543 ± 0.0312	0.7687 ± 0.0298 (RF)
361616	$\hat{y} = 2.776x_4 + x_5(0.860x_3 + 3380.399x_6) - 150.374$	0.9454 ± 0.0048	0.9433 ± 0.0069 (SR)
361617	$\hat{y} = -0.116x_4 \left(-0.098x_2 - 26.977x_6 - \frac{0.307}{x_7 - 0.045} + \frac{0.082}{x_0 - 0.5}\right)$	0.9536 ± 0.0078	0.9988 ± 0.0005 (XGB)
361622	$\hat{y} = -\frac{89.49x_0}{280.988x_{12} - 312.119x_7 + 1043.105} + 4.133 \cdot 10^3(x_1 + x_{14} + x_{16}) + 1.28 \cdot 10^5(x_{10} + x_{12} + x_7) - 639.127e^{-964.96x_1 + \frac{2.818 \cdot 10^3x_9}{\sin(x_2)}} - 832.179 \log(x_0) + 6.088 \cdot 10^4$	0.8909 ± 0.0296	0.9449 ± 0.0193 (SR)

Table 6. Equations designed by a human, assisted by a LLM, starting from the results obtained by SR in a 10-fold cross-validation.

the differences are reduced as soon as the algorithms are given access to a larger amount of computational resources. As predicted by the *no free lunch* theorem [26], SR seems to be inherently more suited for some types of problems (for example, where predictions benefit from a periodic function), while the tree-based ensembles excel in others (for example, ordinal regression tasks).

Still, the results obtained so far seem to indicate that in a few cases SR might be able to obtain even better results with more computational effort and an even vaster search space, so future works will explore even more ambitious hyperparameter settings.

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A A note on this Appendix

The sections of this Appendix are not meant to be a proper part of a scientific paper, yet; they are added to the main text to provide extra information, should the reader be interested in more detailed explanations. The text in the Appendix is still in a draft format, and it will need to be reworked before being considered a proper scientific document. In particular, equations created by SR are often extremely large and sometimes they don't fit inside the page, even when using

solutions such as reducing text size or rotating the page. The text of the main paper already contains all the necessary information to evaluate this work.

B Considerations on the datasets

B.1 `white_wine` and `red_wine`

Inspecting the datasets for the two tasks 361249 (`white_wine`) and 361250 (`red_wine`), on which SR performs significantly worse than the other approaches using both default and enhanced hyperparameters, reveals them to be quite unusual among regression problems. Both datasets present quantitative, real-valued features about wine (such as pH, density, content of sulfur dioxide, etc.), with the target y being an assessment of quality performed by human experts. This quality assessment is represented by an integer, which in dataset `white_wine` ranges from 3 to 9, for a total of only seven different values among the 4,898 samples, and in `red_wine`, the quality label ranges from 3 to 8, for just six different values for 1,599 samples.

It is interesting to notice that the same tasks could be easily reframed as classification problems, as the target $y \in \mathbb{N}^+$; and in fact, the ML domain defines problems of this kind as *ordinal classification* or *ordinal regression* [17], because the relative values of the integer labels have a meaning (e.g., in the case of these datasets, an evaluation of 5 is better than an evaluation of 4). RF and XGBoost employ similar strategies for both regression and classification, thus it makes sense that they might be particularly suited for problems that lie at the edge between the two tasks.

Using a classical MSE-based fitness for ordinal regression might not be the best approach; unfortunately, after a cursory exploration of specialized literature, we could not find any SR fitness function specifically tailored for ordinal regression. We thus decided to experiment with a variation of Poisson likelihood:

$$\mathcal{L}(\hat{y}, y) = \hat{y} - y + y \cdot \ln \left(\frac{y}{\hat{y}} \right) \quad (3)$$

Where $\hat{y}$ is the predicted value and y is the actual label. To avoid numerical issues, in practice we will use $\hat{y}' = \max(\hat{y}, \epsilon)$ instead of $\hat{y}$, with $\epsilon = 10^{-8}$. Still, no improvement has been observed so far.

B.2 `naval_propulsion_plant`

The dataset for task 361247 reveals a situation somehow similar to the one described in subsection B.1; while `naval_propulsion_plant` has 11,934 samples, the target y only features 51 distinct values, with $y \in [0.95, 1.00]$ with increments of 0.001. A comparatively small amount of different values for the target can give an edge to approaches based on DTs, which partition the feature space into a set of regions and predict a constant value within each region. It is also worth mentioning that 361247 is the only task for which PySR with default hyperparameters (less

computational resources, narrower search space) performs significantly *better* than the same algorithm with enhanced hyperparameters. We hypothesize that this is probably due to a premature stop of the evolutionary search: In this particular case, the larger search space (more operators, higher maximum complexity) might be more difficult to explore, especially if SR quickly started considering more complex equations, which are typically slower to evaluate.

In order to test this hypothesis, we rerun the algorithm, increasing the timeout to 5 hours, and the mean performance indeed improved to $R2 = 0.7108 \pm 0.1223$.

B.3 superconductivity

Another dataset with a relatively small number of unique values for the target when compared to the number of samples: 3,007 distinct values over 21,263 samples.

B.4 pumadyn32nh

On the OpenML website, the description of the `pumadyn32nh` dataset used in task 361259⁶ is: “*A realistic simulation of the dynamics of a Puma 560 robot arm. The task in this dataset is to predict the angular acceleration of one of the robot arm’s links. The task is nonlinear and has high noise. The simulator (sic) got as input the angular positions, velocities and torques and other dynamic parameters of the Puma arm. The goal was to predict the acceleration of link 6 θ_{6}''* ”.

This is one of the datasets where SR performs better than the tree-based ensembles, and observing the quite compact equation obtained through the LLM-assisted analysis in subsection E.13, $\hat{y} = c_0 x_{15} \cos(x_4)$, we can hypothesize that it’s probably due to the availability of the transcendental functions among the SR operators, as modeling angular acceleration clearly benefits from functions typically working on angles.

B.5 fifa

The equation found for `fifa` employs a single feature, x_4 , which corresponds to the feature called `overall`. While the dataset description found on the OpenML website does not explicitly describe all features in detail, searching on the internet revealed that `overall` corresponds to the best overall ranking of the player. It thus makes sense that the wage of a player is mostly, if not entirely, explained by their relative ranking.

C Search space of hyperparameter optimization

In the following, we report the (`min`, `max`) range or the list of options [`option1`, `option2`] used during hyperparameter optimization for RF and XGBoost with Optuna [1].

⁶ <http://openml.org/search?type=data&status=active&id=44981>

C.1 Random Forest

```
1 "n_estimators": {  
2     "min": 100,  
3     "max": 5000  
4 },  
5 "max_depth": {  
6     "min": 3,  
7     "max": 30  
8 },  
9 "min_samples_split": {  
10    "min": 2,  
11    "max": 20  
12 },  
13 "min_samples_leaf": {  
14    "min": 1,  
15    "max": 20  
16 },  
17 "max_features": {  
18    "min": 0.1,  
19    "max": 1.0  
20 },  
21 "n_jobs": -1,  
22 "random_state": 42  
23 },
```

C.2 eXtreme Gradient Boosting

```
1 "n_estimators": {  
2     "min": 100,  
3     "max": 5000  
4 },  
5 "max_depth": {  
6     "min": 3,  
7     "max": 30  
8 },  
9 "min_leaves": {  
10    "min": 0,  
11    "max": 64  
12 },  
13 "learning_rate": {  
14    "min": 0.001,  
15    "max": 0.3  
16 },  
17 "subsample": {  
18    "min": 0.5,  
19    "max": 1.0  
20 },
```

```

21 "colsample_bytree": {
22     "min": 0.5,
23     "max": 1.0
24 },
25 "colsample_bylevel": {
26     "min": 0.5,
27     "max": 1.0
28 },
29 "min_child_weight": {
30     "min": 1,
31     "max": 20
32 },
33 "gamma": {
34     "min": 1e-08,
35     "max": 10.0
36 },
37 "reg_alpha": {
38     "min": 1e-08,
39     "max": 10.0
40 },
41 "reg_lambda": {
42     "min": 1e-08,
43     "max": 10.0
44 },
45 "booster": [
46     "gbtree",
47     "dart"
48 ],
49 "grow_policy": [
50     "depthwise",
51     "lossguide"
52 ],
53 "bootstrap": true,
54 "n_jobs": -1,
55 "random_state": 42

```

D Enhanced hyperparameters for PySR

```

1  "niterations": 1000,
2  "population_size": 100,
3  "maxsize": 300,
4  "binary_operators": [
5      "+",
6      "-",
7      "*",
8      "/"
9  ],
10 "unary_operators": [

```

```

11     "sin",
12     "cos",
13     "log",
14     "exp"
15 ],
16 "temp_equation_file": true,
17 "random_state": 42,
18 "procs": null,
19 "parallelism": "multiprocessing",
20 "timeout_in_seconds": 1440

```

E LLM-assisted analysis of SR equations

One of the stated advantages of Symbolic Regression over other types of regressors is the alleged interpretability of the equations. However, the experimental evaluation showed that in order to become competitive with state-of-the-art tree-based models, it is necessary to increase the maximum allowed equation size, add extra operators to the building blocks, and use a considerable extra amount of computational resources. To assess possible loss of interpretability, in this section we will qualitatively analyze selected equations discovered by Symbolic Regression. As we aim to perform this analysis for multiple datasets, we choose to employ the support of a Large Language Model (LLM), the free version of Google Gemini[®] 3.1, accessed between May 19 and June 6, 2026, using the “Pro” mode with the “Extended reasoning” option. After uploading the $\text{\LaTeX}$ equation files to Gemini, we employ the following prompt:

First prompt for the LLM-assisted analysis

Hello Gemini! I am going to pass you several files containing equations in $\text{\LaTeX}$, can you tell me what are the similarities? For example, similarities in structure (same terms) or presence of the same variables (numbered $x_1, \dots, x_n$). I am interested in the common parts of all these equations.

In case the model did not provide a generic equation summarizing all the candidate solution after the first prompt, a second prompt with an explicit request was used:

Second prompt for the LLM-assisted analysis

Going back to the equations, could you create a generic expression catching most of the common parts with undefined constants such as $c_1, c_2, \dots$?

The results of the automatic analysis performed by Gemini are then manually inspected for errors, and used as a base to further develop an expert analysis.

The datasets analyzed in the following were selected according to the following criteria:

1. Either SR performed on par with or significantly better than the other approaches;
2. Or SR's performance achieved $R2 > 0.6$.

E.1 Task 361616, default hyperparameters

In task 361616 (Moneyball), Symbolic Regression already outperforms the tree-based ensembles even with default hyperparameters. The equations selected using the validation set method for each fold are:

$$\hat{y} = x_4 \cdot 2.73 + (x_3 + x_6 \cdot 3.67 \cdot 10^3) (x_5 - 0.0473) - (x_9 \cdot (x_1 - x_9 + 2.30) + 125.) + 5.72 \quad (4)$$

$$\hat{y} = x_4 + x_5 \left(x_3 + \frac{x_6 (x_4 + x_9 + 103.)}{0.0611} \right) - x_8 - \left(\frac{x_1}{x_7} - x_6 x_9 \right) \quad (5)$$

$$\hat{y} = x_2 (-0.432) x_6 - x_3 (-0.261) + (x_4 + x_5 (x_6 \cdot 2.47 \cdot 10^3 - 172.)) 2.52 \quad (6)$$

$$\hat{y} = -0.00388 x_2 x_4 + x_4 \cdot 10.7 + x_5 (x_3 + x_6 \cdot 3.28 \cdot 10^3) - 188. \quad (7)$$

$$\hat{y} = \left(x_4 + \left(-(-2.03) x_3 + x_6 \left(1.86 \cdot 10^4 - \frac{3.47 \cdot 10^3}{x_5} \right) \right) 0.0511 - 60.2 \right) 2.79 \quad (8)$$

$$\hat{y} = (x_3 + x_6 \cdot 4.00 \cdot 10^3) 0.846 x_5 - (146. + x_4 (-2.76)) \quad (9)$$

$$\hat{y} = -\frac{x_1}{0.242} + x_2 (-0.286) + x_5 (x_3 \cdot 0.874 + x_6 \cdot 3.46 \cdot 10^3) - (x_4 + x_9) (-2.71) + 401. \quad (10)$$

$$\hat{y} = -x_1 - x_8 - x_9 (x_1 + x_9 (-0.809)) + (x_3 + x_4 \cdot 9.76 + 577.) (x_5 + x_6 \cdot 0.502 - 0.251) + 128. \quad (11)$$

$$\hat{y} = -x_1 - x_1 - x_1 + x_4 + x_4 + x_5 \cdot 1.92 \cdot 10^3 + x_6 x_6 \cdot 1.31 \cdot 10^3 - (x_3 (-0.310) - x_4 - 1.07) - 584. \quad (12)$$

$$\hat{y} = -x_1 + x_5 (x_5 (-0.955) - x_6 \cdot 2.92) \left(x_2 \cdot 137. \frac{1}{x_3} - 1.55 \cdot 10^3 \right) + (-x_1 + x_4 + x_9) 2.56 - 63.7 \quad (13)$$

The equations are already quite intricate, even with the initial limitation of complexity to 30, and finding common elements without automatic support would require a considerable amount of manual inspection. The text produced by the LLM in response to the prompt in Text Box E claims to have found the following patterns:

1. Variables x_3 , x_4 , x_5 , and x_6 appear in every equation;

2. The majority of equations feature a linear additive term as a function of x_4 , with a constant $c_1 \approx 2.7$;
3. A pattern $x_5 \cdot (x_3 + c_2 \cdot x_6)$ also appears frequently, with c_2 typically very large $c_2 \approx 10^3$.

After concluding the analysis, the LLM suggests the following minimal equation form:

$$\hat{y} \approx c_1 x_4 + x_5(x_3 + c_2 x_6) \quad (14)$$

We test the proposed equation 14, optimizing its constants c_1, c_2 on the whole **Moneyball** dataset using the Nelder-Mead algorithm. The results, however, are not satisfying, with a negative R^2 value. Looking back at the analysis, the LLM did not consider that x_3 could also be multiplied by a constant, and that all equations 4-13 contain a separate constant term. Reformulating the equation as:

$$\hat{y} \approx c_1 x_4 + x_5(c_2 x_3 + c_3 x_6) + c_4 \quad (15)$$

And re-optimizing constants $c_1, \dots, c_4$ again using Nelder-Mead on the whole dataset, this time we obtain $R^2 = 0.9458$ with the final equation being:

$$\hat{y} = 2.776x_4 + x_5(0.860x_3 + 3380.399x_6) - 150.374, \quad R^2 = 0.9458 \quad (16)$$

The performance of this equation is on par with the mean $R^2 = 0.9453$, and even if it appears slightly higher, it is important to consider that the constants were optimized on the whole dataset, without a cross-validation, so overfitting might play a small role. As a final validation, we perform a 10-fold cross-validation using the task's default folds, optimizing the constants on the training set and evaluating the R^2 value on the test, for a final mean $R^2 = 0.9454 \pm 0.0048$.

E.2 Task 361234, default hyperparameters

Task 361234 (**abalone**) is the other example where Symbolic Regression outperforms tree-based ensembles even with default hyperparameters. Here below are the equations found for each fold:

$$\hat{y} = \left(-0.202 + \frac{x_4 + x_7 + 0.0828}{x_5 + 0.0721} \right) \left(-x_6 + 2.88 + \frac{x_3 + x_4 - x_6}{x_1} \right) \quad (17)$$

$$\hat{y} = x_3 + x_3 + 2.48 + \frac{-x_4 - x_4(-0.506) + x_6 - 1.44}{(-1)0.103 + \frac{x_4 + \frac{x_5 + 0.0633}{-0.297}}{20.9x_7}} \quad (18)$$

$$\hat{y} = \frac{x_4(2.34 - x_6) \left(x_2 + x_3 + 0.283 - \frac{0.0357}{x_2} + \frac{x_7}{x_1} \right)}{0.742x_5} + 2.09 \quad (19)$$

$$\hat{y} = -(x_1 - x_5x_4 \cdot 0.779 + x_6 \cdot 7.33) + 3.43 + \frac{\frac{1.87 - x_3}{-x_7 - 0.271} + 7.09}{\frac{1}{x_4}x_5} \quad (20)$$

$$\hat{y} = -x_6 + \left(x_4 + x_7 \cdot 5.04 \frac{1}{x_5 + 0.174} - \left(x_6 - \left(-x_6 + \frac{x_4 - x_7}{x_5} \right) \right) 1.07 - 0.325 \right) 2.49 + 0.129 \quad (21)$$

$$\hat{y} = x_6 + (x_4 - x_6) \left(-(x_7 + x_7) + \frac{(x_3 + x_7)4.38 + 3.65 + \frac{x_6}{x_1}}{x_5 - -0.0228} \right) + 0.709 \quad (22)$$

$$\hat{y} = -x_1 + x_4 - (x_1 - 0.513) + \left(x_3 + \frac{x_4 + x_7 - (x_5 + x_6)}{x_5 + 0.143} \right) 5.90 + 3.01 \quad (23)$$

$$\hat{y} = (-1)(-1.06) + \frac{-x_4 - 1.88}{\frac{1}{-10.3} \left(-x_3 + x_5 + \frac{x_5}{x_7} + x_6 + \frac{x_1 + x_5}{x_4} \right)} \quad (24)$$

$$\hat{y} = -x_2 + x_4 + x_5 + x_7 \cdot 8.41 \frac{1}{x_5 + (x_4 - x_6 + x_7(-1.30))(-0.366) + 0.0676} + 2.84 \quad (25)$$

$$\hat{y} = -6.75 + \frac{4.11(x_3 - (x_6 + 1.17) + (x_4 + x_7 - -0.996)1.63)}{x_5 + 0.202} + \frac{x_5 + x_5}{x_1} \quad (26)$$

The common patterns alleged by the LLM-generated text in response to the prompt in E are:

1. The presence of x_5 in the denominator in each equation;
2. Every equation is built around a single main fractional term multiplied by a constant;
3. The variables located in the numerators (x_4 , x_2 , or x_7) are strictly linear; no variables are raised to a power;

4. Variable frequency: x_5 (10/10), x_4 (7/10), x_7 (3/10), x_2 (2/10).

Furthermore, the text states that the equations could be grouped in three different clusters, featuring canonical forms:

$$\hat{y} \approx c_1 \frac{x_4}{x_5 + c_2} \quad (27)$$

$$\hat{y} \approx c_1 + c_2 \frac{x_7}{x_5 + c_3} \quad (28)$$

$$\hat{y} \approx c_1 \frac{x_2 x_4}{x_5} \quad (29)$$

After a manual inspection, we realize that the LLM text ignores the presence of x_6 (in 10/10 equations!), x_3 (8/10), and x_1 (6/10). Starting from equations 27-29, but also considering x_6 and x_3 we design a new equation:

$$\hat{y} \approx \frac{c_1 x_3 + c_2 x_4 + c_3 x_6 + c_4}{c_5 x_5 + c_6 x_7 + c_7} + c_8 \quad (30)$$

This time, optimizing constants $c_1, \dots, c_8$ with Nelder-Mead on the whole **abalone** dataset gives inconsistent results, so we switch to CMA-ES [11] for evolutionary numerical optimization. The final best result has a $R^2 = 0.5667$, comparable to the original $R^2 = 0.5652$, and higher than XGBoost's at $R^2 = 0.4919$, with final equation:

$$\hat{y} = -3.259 + \frac{19.441x_3 + 44.094x_4 - 34.611x_6 + 7.709}{5.870x_5 - 1.487x_7 + 1.236}, \quad R^2 = 0.5667 \quad (31)$$

The final R^2 value might appear slightly higher, but it is important to notice that for this qualitative study the constants of Eq. 30 have been optimized on the whole dataset, so overfitting might play a role. As a final validation, we perform a 10-fold cross-validation using the task's default folds, optimizing the constants on the training set and evaluating the R^2 value on the test, for a final mean $R^2 = 0.5662 \pm 0.0254$.

E.3 Task 361622, enhanced hyperparameters

In task 361622 (**cars**), Symbolic Regression is already non-significantly different from Random Forest and XGBoost with default hyperparameters; however, with enhanced hyperparameters its mean R^2 value improves significantly, so in the following we report the results for the latter case. As the maximum size for equations is set to 300 and more operands can be used, we expect considerably more complex solutions than in cases with default hyperparameters. The equations selected using the validation set method for each fold are:

$$\begin{aligned} \hat{y} = & - (x_0 \cdot 0.164 + (x_1 - x_7) e^{\cos(x_{15}x_3) + \frac{3.67}{x_2} x_1 \cdot 9.46 + 1.06}) \\ & + \left(x_1 \cdot 3.08 \cdot 10^3 + (x_1 - x_{14}) \left(-(-6.97 \cdot 10^3) x_{10} - x_7 (-4.11 \cdot 10^3) + \cos\left(\frac{x_{13}}{e^{x_9 - 1.51}}\right) (x_1 - \sin(-(x_1 - x_{10}) + e^{x_{11}})) (-195.) + e^{x_1 (\cos(x_{15}) + \frac{e^{x_2}}{0.192x_0 \sin(x_1)})} + 4.41 \cdot 3.60 \right) \right) \\ & + e^{x_6 + 7.03} \end{aligned} \quad (32)$$

$$\begin{aligned} \hat{y} = & \left((1.14x_1 + 3.88x_{13} + 3.88x_7 + 4.08 \sin(x_1x_6 + x_1 + x_{14} - x_{15} - 1.78)) (x_2 + e^{x_{10}} + \sin(\log(x_0)) - \cos(x_{15} - 2.02)) + (x_{11} - x_{15} + \cos(x_1x_9 + x_1 + (-3.69x_6 + x_8 - 6.79) e^{x_{15} \log(x_0)} + \sin((\cos(3.69x_0x_{15} \\ & - (\log(x_0) + 21.7) (-x_{14} + \sin(x_0) + 7.44) e^{(x_{13} - x_5 + x_8)}) e^{-\sin(-(x_1 + x_{12} + x_8 + 0.841))} \right) \end{aligned} \quad (33)$$

$$\begin{aligned} \hat{y} = & - \frac{x_0}{-x_{12} - x_7 + 5.54} \\ & + x_1 (x_{13} - 1.00x_8) (2x_1^2 x_5 (x_1 - 11.8x_6) + 9.43) \\ & - 2x_{13} \\ & - 1.74x_2 (-x_9 + (28.0x_{13} + x_2 + 1.24) (x_{14} - x_{16} - e^{\cos(x_1 - 2.01x_{11})} - 1.19) + 6.72) e^{x_2} \\ & + x_7 (1.47 \cdot 10^4 - x_1) - x_7 \\ & - (-1.18 \cdot 10^3 (x_1 - 3.88) e^{-x_9 - 4.30 \cdot 10^3} e^{x_{12}} \\ & + 434.e^{x_5} - e^{x_1 + x_{15}} \\ & + e^{x_1 + 2.06x_{10}} \\ & - e^{x_1 + x_{15} - x_6} \\ & + \log(x_4 + 0.0433) + 1.02 \cdot 10^4 \end{aligned} \quad (34)$$

$$\begin{aligned} \hat{y} = & x_{10} \cdot 1.35 \cdot 10^4 \\ & + (x_1 + x_{16}) (x_{12} + x_7 + 1.74) \left(\frac{x_0}{-63.7} + (x_1 + x_8) (- (x_{13}x_9 \cdot 80.6 - (-x_{11} + x_{12}) 52.3) - 114.) + 3.05 \cdot 10^3 \right) \\ & - (x_{15} + x_7) (-1.54 \cdot 10^3) \\ & - (x_5 - 3.39) (-651.) \\ & + ((x_{13} + x_6) (x_{11} \cdot 466. + e^{x_1 + 254.}) - 398.) 5.02 \\ & - (x_{12} + x_{12} + x_6) 1.61 \cdot 10^3 \end{aligned} \quad (35)$$

$$\begin{aligned} \hat{y} = & - \frac{x_0}{x_{11} + x_{13} - \sin(x_1 + x_3 (-0.587)) + \sin(x_1 + x_4 - 47.7) + 4.52} \\ & + x_1 (x_1 + x_{11} + (x_{11} + x_8) (x_{13} + \sin(x_1))) (x_1 + x_{12} + x_6) \left(-x_9 - \frac{x_{11} - x_{14} \cdot 3.08}{x_{15} - 0.953} + 47.4 \right) \\ & + ((x_{10} + x_{12} + x_7) 1.01 \cdot 10^3 - 337.) 13.9 \\ & + \left(\frac{x_{12}}{x_7 - 0.384} + x_{16} \right) 2.88 \end{aligned} \quad (36)$$

The text produced by the LLM in response to the first prompt E revealed several common parts:

1. $x_7 + x_{12} + x_{10}$ often appear together with a linear shift;
2. The same holds for $x_1 - x_{14} + x_{16}$;
3. Exponential interaction $e^{x_1 - x_9}$.

The answer to the second prompt produced the following generic equation:

$$\hat{y} = c_0 + c_1(x_1 - x_{14} + x_{16}) + c_2(x_{10} + x_{12} + x_7) - \frac{c_3 x_0}{c_4 + x_{12} + x_7} + c_5 e^{\frac{c_6 x_1 - x_9 + \frac{c_7}{\sin(x_2)}}{\sin(x_2)}} + c_8 \log(x_0) \quad (42)$$

After a first attempt at constant optimization using CMA-ES [11], a few extra constants were added to multiply x_{12} and x_7 when they appear at the denominator:

$$\hat{y} = c_0 + c_1(x_1 - x_{14} + x_{16}) + c_2(x_{10} + x_{12} + x_7) - \frac{c_3 x_0}{c_4 + \mathbf{c}_9 x_{12} + \mathbf{c}_{10} x_7} + c_5 e^{\frac{c_6 x_1 - x_9 + \frac{c_7}{\sin(x_2)}}{\sin(x_2)}} + c_8 \log(x_0) \quad (43)$$

Final equation, with optimized constants:

$$\begin{aligned} \hat{y} = & -\frac{89.49x_0}{-280.988x_{12} - 312.119x_7 + 1043.105} + 4133.879x_1 + 12806.192x_{10} + \\ & 12806.192x_{12} + 4133.879x_{14} + 4133.879x_{16} + 12806.192x_7 \\ & - 639.127e^{-964.96x_1 + \frac{2818.088x_9}{\sin(x_2)}} - 832.179 \log(x_0) + 6088.817 \\ R^2 = & 0.8990 \end{aligned} \quad (44)$$

The value $R^2 = 0.8990$ is lower than the original mean $R^2 = 0.9449$, but it can still be considered as a high-quality solution; and on the other hand, the equation is considerably simpler than the equations in 32-41. A 10-fold cross-validation shows that optimizing the equation's constant on the whole dataset only leads to a modicum amount of overfitting, with mean $R^2 = 0.8909 \pm 0.0296$.

E.4 Task 361264, default hyperparameters

Task 361264 (**socmob**) is a peculiar case, where no algorithm performs significantly better than the others, probably due to a large variance between folds; even using enhanced hyperparameters or hyperparameter tuning does not seem to significantly improve performance, so for this case we select the equations that PySR discovered with its default hyperparameters.

$$\hat{y} = x_4 x_4 (-0.00122) + x_4 \cdot 1.09 - (x_2 - 1.45) - \left(\frac{x_4}{7.94 - x_1} + \frac{x_4}{0.733 - x_1 x_3} \right) 0.171 \quad (45)$$

$$\hat{y} = -\frac{x_3}{-x_2 - 0.111} + 0.875 \left(x_4 + \frac{0.443}{(x_1 - 4.13) \frac{x_4}{4.66 - (x_1 - 3.38)} - 0.537} \right) \quad (46)$$

$$\hat{y} = x_4 + \frac{x_4 \left(0.711 - \frac{0.850}{x_1 + x_1 - 7.90} \right)}{\frac{1}{(-1)1.74 \frac{1}{x_1 - 0.573}} (x_1 - 8.05)} + 1.20 \quad (47)$$

$$\hat{y} = x_3 \left(2.76 - x_4 (-0.167) \frac{1}{x_1 - 7.93} \right) + \frac{x_4 \left(1.27 + \frac{0.310}{x_1 - 0.598} - \frac{0.486}{3.55 - x_1} \right)}{1.43} \quad (48)$$

$$\hat{y} = - (x_3 (-(-9.98) x_2 - 10.4) - x_4 - 65.8) - \frac{5.36 \cdot 10^4}{821. - x_4} \quad (49)$$

$$\hat{y} = x_3 + \frac{x_3}{(x_1 - 0.562) (x_2 + 0.0690)} + x_4 \left(-1.00 x_4 + x_4 - -1.03 + \frac{0.309}{x_1 - 7.87} \right) \quad (50)$$

$$\hat{y} = (5.99 - x_2) (x_3 + x_4 (0.162 - x_4 x_4 \cdot 2.47 \cdot 10^{-7})) \quad (51)$$

$$\hat{y} = x_4 \frac{1}{x_1 - 4.10} \cdot 271. \frac{1}{x_4 \cdot 0.185 + (x_1 - 8.02) (-798.)} - (x_2 - x_3) 2.62 + (x_4 + 3.60) 0.910 \quad (52)$$

$$\hat{y} = -1.46 \cdot 10^{-6} x_4 x_4 x_4 + x_4 + x_4 \cdot 0.216 \frac{1}{x_1 - 7.89} \quad (53)$$

$$\hat{y} = x_4 \left(1.16 - 0.00133 \left(x_4 - \frac{7.28}{8.01 - x_1} + \frac{47.1}{0.194 - x_1} \right) \right) \quad (54)$$

The text produced by the LLM in response to the prompt in E highlights the following points:

1. x_4 is the primary driver, often appearing in the form $c \cdot x_4$;
2. x_1 often appears as the denominator of a fraction in the form $\frac{1}{x_1 - c}$, with two recurring values $c_1 \approx 0.8$ and $c_2 \approx 0.6$

3. x_2 and x_3 often appear in linear terms.

After a deeper human analysis, we noticed that the equations for 3 folds were just functions of x_1, x_4 . Thus, we tried a first equation:

$$\hat{y} \approx c_1 x_4 + \frac{c_2 x_4}{x_1 - c_3} \quad (55)$$

Optimizing the constants c_1, c_2, c_3 using CMA-ES produced the following equation:

$$\hat{y} = 0.896x_4 - \frac{0.358x_4}{x_1 - 3.422}, \quad R2 = 0.7536 \quad (56)$$

Assessing the overfitting with a 10-fold cross-validation, we obtained a mean $R2 = 0.7299 \pm 0.1045$, which is close to the original value of $R2 = 0.7661 \pm 0.0908$ found by Symbolic Regression over the more complex Eqs. 45-54. Adding another linear term in x_3 to Eq. 55:

$$\hat{y} \approx c_1 x_4 + \frac{c_2 x_4}{x_1 - c_3} + c_4 x_3 \quad (57)$$

And optimizing again the constants with CMA-ES, produced the following equation

$$\hat{y} = 6.325x_3 + 0.936x_4 - \frac{0.24x_4}{x_1 - 3.701}, \quad R2 = 0.7813 \quad (58)$$

Optimizing the constants in a 10-fold cross-validation gives a mean $R2 = 0.7597 \pm 0.0859$. Adding a further linear term $c_5 x_2$ produces no further improvements.

E.5 Task 361272, default hyperparameters

In task 361272 (`fifa`), Symbolic Regression's performance is not significantly different from the other algorithms, already with default hyperparameters. The equations identified by PySR are:

$$\hat{y} = -x_3 + ((x_4 - 55.6)(x_4 + x_4 - 134.)(-0.463x_{15} + x_{15} + x_4 + x_4 + x_4 - 201.) + 2.79 \cdot 10^3) 1.13 \quad (59)$$

$$\hat{y} = (x_4 - 51.8)(x_8 + (x_4 - 64.4)(-x_0 + x_4(-x_6 - 38.9)(71.4 - x_4)0.000881 + x_5) + 171.) \quad (60)$$

$$\hat{y} = (x_4 - 53.6)x_5(x_4 - 53.6)0.0134(28.0 - (6.35 - x_9)(x_4 - 63.2)(x_4 - 64.5)0.00249) \quad (61)$$

$$\hat{y} = (x_8 + (x_4 - 55.5)(54.8 - x_4)(-3.18))((x_4 - 60.1)(x_4 - 66.8)x_9 \cdot 0.000894 + 9.77) \quad (62)$$

$$\hat{y} = (x_4 (x_4 \cdot 7.06 - 912.) + 2.96 \cdot 10^4) (x_4 + x_6 \cdot 0.139 - 56.5) \quad (63)$$

$$\hat{y} = (x_4 (x_4 x_4 \cdot 0.0359 - 443.) + 1.91 \cdot 10^4) \left(-\frac{x_{27}}{x_6} + x_4 - 51.1 \right) \quad (64)$$

$$\hat{y} = \left(x_{15} + \frac{x_4 - 61.3}{0.145} \right) (x_4 - 64.9) \left(x_4 - 60.5 - \frac{-x_0 + \frac{x_{14}}{x_{22}} + 22.7}{\frac{1}{-17.1} x_{15}} \right) - 2.37 \cdot 10^3 \quad (65)$$

$$\hat{y} = x_4 (-x_0 (x_0 \cdot 0.593 - 29.6) + x_{15} + x_4 (x_4 \cdot 7.89 - 1.43 \cdot 10^3) + 8.53 \cdot 10^4) - (x_{13} - 0.111) - 1.71 \cdot 10^6 \quad (66)$$

$$\hat{y} = -x_3 + (x_4 - 51.9) ((-x_0 + x_2) 3.93 + (x_4 - 64.5) (x_{15} - x_4 (-7.30) - 531.)) \quad (67)$$

$$\hat{y} = (x_4 - 51.1) (x_4 - 62.9) (x_4 - 62.9) 7.11 \quad (68)$$

The text produced by the LLM in response to the prompt claims the following similarities:

1. x_4 is the only variable that appears in every single equation across all 10 folds;
2. The most common mathematical pattern is the use of binomial offsets in the form of $(x_4 - C)$, with $C \approx 60$;
3. The common structural blueprint here relies on deeply nested multiplicative blocks;
4. Other variables have low frequency of appearance, with x_{15} (3/10 folds) being the most common.

In response to the second prompt, the LLM produces the following equation:

$$\hat{y} \approx -c_1 x_3 + c_2 (x_4 - c_3)^n \left[c_4 x_4 (c_5 x_4 - c_6) + c_7 x_k + c_8 \right] \quad (69)$$

with x_k which could be x_{15} , x_0 , or x_6 . The equation, however, contradicts some of the previous statements in the text. After a manual review, we settle on the tentative expression:

$$\hat{y} \approx c_0 (x_4 - c_1) (x_4 - c_2) (x_4 - c_3) (x_4 - c_4) + c_5 \quad (70)$$

Optimizing the constants with CMA-ES provides the following equation:

$$\hat{y} = 0.17 (x_4 - 56.201)^4 + 1823.898, \quad R2 = 0.7568 \quad (71)$$

Which is a somehow surprising result, with constants $c_1 \approx c_2 \approx c_3 \approx c_4$. Optimizing the constants with a 10-fold cross-validation delivers a mean $R2 = 0.7543 \pm 0.0312$, indicating only a small amount of overfitting.

E.6 Task 361257, default hyperparameters

In task 361257 (**diamonds**), no algorithm is significantly better or worse than others, mainly due to a large standard deviation in performance over the folds. Considering Symbolic Regression with default hyperparameters, the equations obtained are:

$$\hat{y} = (x_6 - 3.35) (x_0 (x_0 (-108.) + x_3 x_4 (x_2 (-1.00) + x_6) + 1.24 \cdot 10^3) - 23.8) \quad (72)$$

$$\hat{y} = -x_1 - x_6 (x_0 (x_0 (x_3 (x_2 (x_3 + 22.0) - 187.) + x_8) + x_1 (-24.7) - 621.) + 376.) + 1.13 \cdot 10^3 \quad (73)$$

$$\hat{y} = x_3 + (x_3 + x_6 (x_0 (x_1 - -55.0) - 51.6) + 151.) (x_3 (-x_2 + x_6 - 0.358) - x_6 + 21.8) + 473. \quad (74)$$

$$\hat{y} = ((x_7 - 2.16) (x_0 \cdot 65.6 - 35.8) - -54.7) (-0.291x_1 + x_1 - x_3 (x_2 - x_7) - x_6 - x_6 + 29.7) \quad (75)$$

$$\hat{y} = x_0 (x_3 + x_7) (x_2 (-47.8) - -961.) - 2.00 \cdot 10^3 \quad (76)$$

$$\hat{y} = x_0 ((-x_2 + x_7) 199.x_3 - -6.19 \cdot 10^3) - 2.41 \cdot 10^3 \quad (77)$$

$$\hat{y} = (x_0 - 0.278) (x_0 x_3 (x_2 (-201.) - -1.58 \cdot 10^3) - -4.94 \cdot 10^3) \quad (78)$$

$$\hat{y} = x_0 (x_6 - 3.82) (x_3 (514. + x_2 (-68.5)) + 1.18 \cdot 10^3) \quad (79)$$

$$\hat{y} = (x_0 - 0.315) (x_6 - 1.79) (x_3 x_4 (-x_2 + x_7) + 814.) + 652. \quad (80)$$

$$\hat{y} = (x_0 \cdot 287. - 88.9) (x_3 (-x_2 + x_7) + 19.2) \quad (81)$$

The text answering the prompt in E lists the following considerations:

1. While not universally present, x_6 and x_7 frequently appear, and they seem to perform a highly similar—or interchangeable—role within the models;

2. In several equations, x_0 acts as a direct, standalone multiplier for the entire core expression, taking the form $x_0 \times (\dots)$;
3. In others, the algorithm applies a simple affine shift to x_0 before multiplying it against the rest of the terms, creating patterns like $(x_0 - 0.278)$ or $(x_0 \cdot 287. - 88.9)$;
4. x_3 modulates an inverse relationship (or subtraction) between x_2 and either x_6 or x_7 ;

The text then proposes the following common skeleton:

$$\hat{y} \approx f(x_0) \cdot [x_3 \cdot (x_{6|7} - x_2) + \dots] \quad (82)$$

After a human intervention, the equation has the new structure:

$$\hat{y} \approx (x_0 - c_0) \cdot (c_1 x_3 \cdot (c_2 x_6 - c_3 x_2)) \quad (83)$$

Optimizing the constants using CMA-ES provides the following equation:

$$\hat{y} = 14.15x_3 (x_0 - 0.305) (-22.734x_2 + 38.959x_6), \quad R^2 = 0.7191 \quad (84)$$

In a 10-fold cross-validation, we obtain a mean value of $R^2 = 0.7191 \pm 0.0068$, with no indication of overfitting.

However, this was one of the few cases where the default choice of PySR was giving a higher mean R^2 . What happens if we do the same with the other set of equations?

1. x_0 , x_2 , and x_3 are present in all 10 equations. This indicates they have the highest feature importance for your target variable
2. x_6 and x_7 are secondary but highly prevalent, appearing in 9 out of the 10 folds
3. x_4 is rarely selected, appearing only in three folds
4. form of $x_3(-x_2 + \text{variable}) + \text{constant}$ or $x_3(C_1 + x_2 \cdot C_2) + C_3$
5. x_0 is a multiplier

$$\hat{y} \approx (x) = (c_1 x_0 + c_2) \cdot (x_j + c_3) \cdot [c_4 x_3 (c_5 x_2 + c_6 x_k + c_7) + c_8] \quad (85)$$

After constant optimization with CMA-ES:

$$\hat{y} = (21.035x_0 - 3.686) (x_6 - 1.788) (0.605x_3 (-4.107x_2 + 3.456x_6 + 4.696) + 40.006), \quad R^2 = 0.9618 \quad (86)$$

Optimizing the constants in a 10-fold cross-validation returns a mean $R^2 = 0.9618 \pm 0.0023$, indicating little to no overfitting.

E.7 Task 361254, default hyperparameters

In task 361254 (`sarcos`), SR performs worse than the tree-based ensembles, but its mean $R^2 = 0.9272$ is quite high and not that distant from the best, XGBoost with $R^2 = 0.9759$. In this case, we consider the equations discovered by PySR with default hyperparameters, as the gain with enhanced hyperparameters is relatively small:

$$y = -x_{10} + x_{14}x_2 + (38.6 + x_{20}(-0.320))\left(x_0 + \frac{x_{17}}{56.9}\right) + \frac{x_{10}x_8 + x_{14} + x_{14} + 16.2}{0.785} \quad (87)$$

$$y = -x_{10} + (x_0 \cdot 14.6 + x_{14} + 7.63)2.58 + \frac{x_0x_{16} - x_{10}x_7 + x_{17}}{1.60 - x_6} \quad (88)$$

$$y = x_0 \cdot 36.7 + x_{14} + \left(x_{18}(-0.164) - x_6 - -11.1 + \frac{-x_{10} + x_{17} + \frac{x_{14} + x_3 + 3.21}{0.462}}{x_3}\right)1.15 \quad (89)$$

$$y = x_0 \cdot 38.0 - x_{18} \cdot 0.171 - \frac{x_7x_8}{0.352} + \frac{x_{14} + \frac{x_{14} + x_{14} + x_{17}}{x_3}}{0.864} - 0.0428 - -20.2 \quad (90)$$

$$y = -x_{10} + x_{12}x_{13} + x_{14} + \frac{x_{17}}{x_3} + (x_0 + 1.71)(x_{14} - (x_8 \cdot 2.02x_7 - 38.1)) - 40.1 - 4.27 \quad (91)$$

$$y = \frac{x_0}{0.0272} - x_{10} - x_{14}(-x_2 - 0.931) + x_{14} - x_8x_7 \cdot 2.89 - 2.00 + 21.8 + \frac{x_{14} + x_{17}}{x_3} \quad (92)$$

$$y = x_0 \cdot 36.1 - x_{10} - x_3(x_6 + 5.37) + (1.81 - x_3 \cdot 0.722)(x_{14} + x_{14} + x_{17}) - \frac{-x_{14} - 23.4}{0.810} \quad (93)$$

$$y = -(-x_0 + x_{10}) + \frac{-x_0x_3(-(-0.135)x_{20} - 8.72) + x_{14} + x_{17} \cdot 0.286 + x_4 + x_5 + 8.26}{0.396} \quad (94)$$

$$y = x_{14}(-35.2) + x_{17}(x_6 - -0.485) + (x_0 + x_{14})(x_2 + 37.8) - (x_7x_8 - x_7(0.601 - x_{10})) - -19.2 \quad (95)$$

$$y = -x_{10} - \left(-\frac{x_6}{-0.281} + \frac{-(x_{14} + x_{14} + x_{17}) - 11.4}{x_3} \right) + (x_1 + x_{14} + (28.4 - x_0)(x_0 + 0.385)) 1.25 \quad (96)$$

The text produced by the LLM answering the prompt in Textbox E lists the following common elements:

1. x_0, x_{14}, x_{17} appear in all equations;
2. The term $C \cdot x_0$ appears often, with $C \in [36, 39]$;
3. $-x_{10}$ appears in 9/10 equations;
4. There is a recurring nonlinear sub-structure where x_{14} and x_{17} are added together, often to form a numerator, and frequently divided by x_3 ;
5. When x_7 and x_8 appear, they generally do so as a multiplied pair (x_7x_8);

The text then presents the following generic equation:

$$\hat{y} \approx 37.5x_0 - x_{10} + \alpha x_{14} + \beta \left(\frac{2x_{14} + x_{17}}{x_3} \right) - \gamma(x_7x_8) + C \quad (97)$$

After a first human intervention, mainly focused on adding more constants to be optimized, the candidate equation becomes:

$$\hat{y} \approx c_0x_0 - x_{10} + c_1x_{14} + \left(\frac{c_2x_{14} + c_3x_{17}}{x_3} \right) - c_4x_7x_8 + c_5 \quad (98)$$

Optimizing the constants using CMA-ES on the whole dataset gives:

$$\hat{y} = 37.795x_0 - x_{10} + 1.149x_{14} - 2.975x_7x_8 + 20.163 + \frac{2.273x_{14} + 1.098x_{17}}{x_3}, \quad R2 = 0.9313 \quad (99)$$

What is remarkable in Eq. 99 is that the constant $c_0 = 37.795$, which multiplies x_0 , indeed has a value in the interval $[36, 39]$, as previously stated by the text. The $R2$ value is also quite high; optimizing the constant values in a cross-validation gives a mean $R2 = 0.9312 \pm 0.0026$, indicating a minimal amount of overfitting.

E.8 Task 361256, default hyperparameters

In task 361256 (`cpu_activity`), SR with default hyperparameters performs significantly worse than RF and XGBoost, but the difference in mean $R2$ is quite small, with $R2_{SR} = 0.9713$ and $R2_{XGB} = 0.9815$. We thus decided to analyze the equations obtained with default hyperparameters, as they tend to be smaller in size:

$$y = \frac{\frac{x_{17} - 1.71(x_{15} + x_{18})}{-39.0} - 75.1}{x_2 \cdot 17.5 \frac{1}{x_{20}} + 0.769} \quad (100)$$

$$y = x_{15}(-0.0627) - x_{17} \cdot 0.0227 + (x_2 + (x_0 + x_{16}) 7.55)(-0.00176) - 97.9 - \frac{9.08 \cdot 10^3}{x_{20} + x_5 + 90.3} \quad (101)$$

$$y = \left(-7.60 \cdot 10^{-6} + \frac{109.}{x_2(x_6 + 25.2) + x_{20} - (x_{15} + x_{17})(-477.)} \right) \left(x_{19}(-3.25) \frac{1}{-0.461x_{14} - 0.334} + x_{20} \right) \quad (102)$$

$$y = -(-0.0224) \left(x_{15}(-2.89) - x_{17} + x_2(-0.0827) + \frac{x_5}{-0.0348} \right) + 97.4 - \frac{1.65 \cdot 10^3}{\frac{1}{5.75}(x_{20} + x_5 - -95.0)} \quad (103)$$

$$y = \frac{x_0(-0.0229) - x_{16} \cdot 0.0432 + x_5(-2.29) + (x_{15} + x_9)(-0.135) + 171.}{x_2 \frac{1}{x_{20}} \cdot 46.9 + 1.75} \quad (104)$$

$$y = \frac{x_{20}}{-\frac{x_0}{-0.419} + \frac{x_2 - \frac{x_{20}}{x_5 - 37.4}}{2.50} + (x_{10} + x_{13} + x_{15} - x_5 - 77.7) 8.80} + 0.947 \quad (105)$$

$$y = \frac{(0.532 + x_{17}(-0.000157)) 167. + \left(\frac{x_{16}}{0.532} + x_4 \right)(-0.00387)}{0.948 - \left(-x_2 + \frac{86.0 - x_{15}}{0.0332} \right) \frac{1}{x_{20}} \cdot 23.2} \quad (106)$$

$$y = 0.0856 + \frac{53.2 - x_5}{0.547 + (x_{15} \cdot 41.9 + x_2 + (x_0 + x_{17})(x_{18} + 3.95)) 13.3 \frac{1}{x_{20}}} \quad (107)$$

$$y = \frac{x_{17} + \frac{x_{15} + x_9}{0.427} - \left(-0.0643x_2 + \frac{x_4 x_6}{-29.5} \right)}{-37.2} + 97.3 - \frac{9.39 \cdot 10^3}{x_{20} \cdot 1.10 + 94.0} \quad (108)$$

$$y = -0.233(x_{10} \cdot 0.233 + x_6 - 0.724) - \frac{2.22}{(-0.244) \left(0.100 - \left(x_{17} + \left(\frac{x_2}{-56.0} - -64.5 \right)(-3.73) \right)(-42.8) \frac{1}{x_{20}} \right)} \quad (109)$$

The text produced by the LLM following the prompt in Textbox E claims to have found the following similarities:

- The variables x_2 and x_{20} are present in every single equation;

- The variable x_{15} appears in 9 out of 10 folds. The variable x_{17} appears in 8 out of 10 folds;
- The variable x_5 appears in 5 folds. The variable x_0 appears in 4 folds. The rest ($x_4, x_6, x_9, x_{10}, x_{13}, x_{14}, x_{16}, x_{18}, x_{19}$) appear sporadically, likely acting as fine-tuning parameters or fitting noise for their specific cross-validation folds;
- In the majority of the folds, x_{20} appears in a denominator, very frequently coupled with x_2 via multiplication;
- x_{15} and x_{17} consistently act as linear drivers.

The candidate equation summarizing the considerations is:

$$\hat{y} \approx (c_1 x_{15} + c_2 x_{17}) + \frac{\text{Numerator}}{c_3 x_2 \frac{1}{x_{20}} + c_4} \quad (110)$$

Interestingly, the Numerator part is not explicit, but we can gather from the context that it is likely meant to be either a constant or a function of the other sporadically appearing variables. After a first human intervention, the equation becomes:

$$\hat{y} \approx (c_0 x_{15} + c_1 x_{17}) + \frac{c_2}{c_3 x_2 + \frac{1}{x_{20}} + c_4} \quad (111)$$

Optimizing the constants using CMA-ES gives the following equation:

$$\hat{y} = -0.058x_{15} - 0.03x_{17} + \frac{0.633}{1.354 \cdot 10^{-7}x_2 + \frac{1}{x_{20}} + 0.006}, \quad R2 = 0.9718 \quad (112)$$

Several interesting details emerge from Eq. 112. First, the contribution of x_2 seems to be extremely small, with the constant set at a value $c_3 \approx 10^{-7}$. Then, the $R2$ value is basically at the same level as the original. Evaluating the new equation in a 10-fold cross-validation gives a mean 0.9708 ± 0.0068 , indicating a small amount of overfitting. Interestingly, removing the term $c_3 x_2$ from Eq. 111 and optimizing the constants again in a cross-validation gives a mean $R2 = 0.9520 \pm 0.0105$, showing that the role of x_2 is not negligible.

E.9 Task 361617, default hyperparameters

In task 361617 (`energy_efficiency`), while SR has the worst result, with default hyperparameters its mean $R2 = 0.9547$ is already extremely high, with XGBoost showing the best performance at $R2 = 0.9988$. While SR's performance increases with enhanced hyperparameters, we decided to analyze the equations obtained with default hyperparameters, as the difference in performance is relatively small, and these equations tend to be smaller in size:

$$y = x_4 x_6 (7.91 + x_6 (-8.47)) + (x_4 - 2.71) (x_2 (x_3 \cdot 0.000217 - 0.00866) - x_3 \cdot 0.0579 + 5.02) + 5.58 \quad (113)$$

$$y = \frac{x_4(x_2 \cdot 0.00217 + x_6 + 0.563) - \frac{0.0226}{0.217 - (x_0 - 0.586)}}{0.298} - 4.16 + \frac{0.0102}{x_6 - 0.00260} \quad (114)$$

$$y = \frac{x_0(0.0150x_2x_6 - 4.31) + x_4}{0.160} + 4.15 - \frac{0.0883}{x_0 - 0.806} - \frac{0.812}{x_3(0.0959 - x_6)} \quad (115)$$

$$y = (x_4(x_6 + 0.263) + x_4) \left(x_2 \cdot 0.00681 - 1.09 - \frac{0.0665}{x_7 - -0.100} \right) - 1.84 - 1.13 \quad (116)$$

$$y = x_4 \left(x_2 \cdot 0.00378 + x_6 - \frac{0.166}{x_6(-x_1x_6 + x_3 + x_4) - -0.707} \right) 3.14 - 1.61 \quad (117)$$

$$y = x_2(-0.0333) + (1.83 - x_0)1.97 - (x_4 - -1.05) \left(-(x_2 \cdot 0.0132 + x_6 \cdot 2.73) - \frac{0.0730}{x_7 - 0.110} \right) \quad (118)$$

$$y = x_6 + \left(-x_0 + x_4 + \frac{5.41}{x_2 \cdot 0.440 - x_3} \right) (x_2 \cdot 0.00919 + x_6 \cdot 4.38 - -0.838) \quad (119)$$

$$y = -x_0 + (0.840 - x_4)(x_2(-0.00721) - (x_6 \cdot 3.94 - -1.90)) - \frac{0.0838}{x_0 - 0.803} - \frac{0.607}{0.147 - x_7} \quad (120)$$

$$y = x_2(x_4 - (2.01 - x_4x_6))0.0113 - 5.63 - \frac{14.6}{-0.414x_2 + x_3 - x_4} \quad (121)$$

$$y = -0.00800(-x_4 - 1.06) \left(x_2(0.140x_4 + x_6) + \frac{3.14}{x_7 - 0.0373} \right) - 5.52 \quad (122)$$

The text generated by the LLM in response to the prompt in Textbox E claims the following common parts:

1. x_2, x_4, x_6 : These three variables are universally present in 100% of the folds;
2. x_0, x_3 : These appear in exactly 50% of the folds (5 out of 10);
3. x_7 : Present in 40% of the folds;

4. Across almost every single fold, x_2 appears with an incredibly small coefficient, usually on the order of 10^{-2} to 10^{-4} ;
5. There is a massive structural consensus that x_4 and x_6 interact multiplicatively, often acting together as the primary slope driver of the equation;
6. Whenever x_0 is used in a fraction, a mathematical pole (singularity) exists precisely around 0.80, with terms like $\frac{1}{x_0 - 0.80}$;
7. Whenever x_7 is selected by a fold, it is exclusively placed in the denominator of a localized rational term.

The equation proposed at the end of the text is:

$$\hat{y} \approx \beta_1 \cdot x_4(\epsilon x_2 + x_6) + \frac{\beta_2}{x_0 - 0.80} + \text{Correction}(x_3, x_7) + C \quad (123)$$

The text states that ϵ is a tiny positive scaling coefficient (≈ 0.001) and β_1 is a primary macro weight. Despite the claims of the last equation, a manual inspection of Eqs. 113-122 reveals that x_4 often multiplies the substructures function of x_0 and x_7 . After a human intervention, the new equation becomes:

$$\hat{y} \approx c_0 x_4 \left(c_1 x_2 + c_3 x_6 + \frac{c_4}{x_0 - c_5} + \frac{c_6}{x_7 - c_7} \right) \quad (124)$$

Optimizing the constants on the whole dataset with CMA-ES gives:

$$\hat{y} = -0.116 x_4 \left(-0.098 x_2 - 26.977 x_6 - \frac{0.307}{x_7 - 0.045} + \frac{0.082}{x_0 - 0.8} \right), \quad R2 = 0.9543 \quad (125)$$

Interestingly, the substructure $\frac{c_4}{x_0 - c_5}$ indeed features $c_5 = 0.8$. The $R2$ value is extremely close as the original, and a 10-fold cross-validation returns a mean $R2 = 0.9536 \pm 0.0078$, revealing only a small amount of overfitting.

E.10 Task 361236, enhanced hyperparameters

In task 361236 (`auction_verification`), SR performs significantly worse than the other algorithms, but still obtains a mean $R2 = 0.9588$, not far from the best result of $R2 = 0.9983$ by XGBoost. The equations obtained by SR during the cross-validation are:

$$y = x_4 + (x_2 + \cos(x_0)) \left(-x_0 \left(\frac{1.49 \cdot 10^3 x_2 e^{\cos(x_6)}}{x_4 (-1.80 e^{\cos(9.47 x_4)} + 2.17 e^{-x_5}) - 0.990} + 0.209 x_4 (x_4 - 1.74) e^{-x_2} e^{\sin(1.78 x_4)} \right) \right) \quad (126)$$

$$y = x_5 + \left(\left((-43.6 \sin(x_0 - x_2 + 2x_5 + 3.70) + 40.5 \cos(x_2 x_4 x_6^2)) \left(4.26 x_3 - 4.26 \sin(x_4 + 0.697) + 4.26 \sin(2x_4) \right) \right) \right) \quad (127)$$

$$y = -x_2(x_2 + 0.544) - x_5 + \left(e^{(\sin(x_0) + \sin(x_0 + 0.226) - 2.23) \sin(-x_5 + \sin(x_5 - 0.734)) + e^{x_3}} + e^{x_2(\sin(x_0) + \sin(x_0 + 0.129))} \right) \quad (128)$$

$$y = x_0 \left((x_6 + x_6) \left(x_2(-14.0) \frac{1}{-x_3 - x_6(-x_2 + x_3 - x_5) - 0.988} + 14.9 \right) + \left(-(-3.79)x_4e^{x_3} + \left(103. - \left(x_5 \right. \right. \right) \right) \quad (129)$$

$$y = (x_2 + \sin(x_5))(x_3 + 3.33)(x_2 \cos(x_5) + x_2 \cos(x_5(x_2 - 0.181) + 130.) + x_2 + 0.495) 1.36 \left(x_0 \left(x_0 x_0(-2) \right. \right) \quad (130)$$

$$y = -x_0 - (-x_3 - x_3 - 0.306) + e^{x_2 + x_5 - x_6 - (0.302 - x_3)} - \frac{\left(x_4 - \left(-x_0 - \frac{x_4(-1.7)}{(x_4x_5 - ((x_6 + 0.224) - \frac{123.}{0.506x_4})22.8)e^{x_2 + x_5 - x_6 - (0.302 - x_3)}} \right) \right)} \quad (131)$$

$$y = \left(\frac{x_0 \left(2.20x_3 + e^{x_0 + e^{0.630 \sin(x_5)}} - \sin(\cos(x_4)) \right)}{x_0 e^{x_0 x_5} e^{-e^{x_3} + \log(x_4) \sin(4.37x_5)} - 0.694} + 1.46 \right) \left(-0.944x_5e^{x_6} + (x_2 - 0.290) \left(x_6 + e^{3.67 \sin(6.61x_5)} \right) \right) \quad (132)$$

$$y = \frac{x_4 \left(x_5 - \left(\frac{x_3 + \sin \left(x_5 + e^{\frac{x_1}{x_1 + x_5 - \sin(x_6) + 0.115}} + e^{\cos(x_5 \cdot 1.08) - 0.0960} \right)}{-0.365} + 4.74 \right) + 1.99 \right) e^{\frac{\sin \left(\frac{\sin(x_0)}{0.468} \right)}{0.433}} + e^{(\sin(x_5) + \cos(x_5))}}{0.459e^{\frac{1}{x_5 + \cos((x_4 + x_4)(-14.2) - 3.41)} \left(x_2(-3.37) + \frac{-x_2 + x_4}{x_5((x_4 - x_6x_6 + x_6 + 2.90)0.619 + 0.768) + \sin \left(\frac{x_4}{1.89} \right) - 1.56} + x_4 \right)}}} \quad (133)$$

$$y = \left(x_3 \sin(x_0) 0.329 e^{\sin((x_3+x_5)2.96)} + \sin(x_0 + 0.0397) \right) \left(x_1 + e^{x_5} (-35.8) - e^{x_2 \sin(-x_2+x_5+0.433)2.57 \frac{1}{0.613}} - \sin(x_2+x_5) \right) \quad (134)$$

$$y = x_1 + \frac{\left(\frac{x_0 - 1.86}{x_4 + 0.579} + \sin(x_0) \right) \left(\left(-x_0 + 7.46 + \frac{2.21}{x_5 - 3.77} \right) x_2 \cdot 15.4 \left(x_3 (9.15 - x_0) - \frac{3.26}{x_5 x_6 (x_2 - x_6) - 0.953} \right) + e^{x_2 - x_5} \right)}{\quad} \quad (135)$$

Answering the prompt in Textbox E, the text produced by the LLM claims the following structural similarities:

1. The variables x_0, x_2, x_3, x_4, x_5 , and x_6 appear in every single equation;
2. The variable x_1 is notably absent from several folds (specifically folds 0, 3, 4, and 6);
3. Every equation makes extensive use of the sin function;
4. The exponential function (e^x) is a universal structural component across the dataset;
5. The log function appears occasionally (e.g., in folds 0, 1, and 6), but unlike the trigonometric and exponential operators, it is not a universally required building block.

The LLM this time did not produce a generic form for the equation at the end of the first prompt, so the second prompt was used. The text produced by the LLM describes a complex structure:

$$\hat{y} \approx \mathcal{L}(\mathbf{x}) + \mathcal{M}(\mathbf{x}) \cdot \psi(\mathbf{x}) + c_1 e^{\mathcal{E}(\mathbf{x})} + \frac{\mathcal{N}(\mathbf{x})}{\mathcal{D}(\mathbf{x}) - c_2} \quad (136)$$

with:

$$\mathcal{L}(\mathbf{x}) = c_3 x_0 + c_4 x_2 + c_5 x_3 + c_6 x_4 + c_7 x_5 + c_8 \quad (137)$$

$$\mathcal{M}(\mathbf{x}) = \sin(c_9 x_0 + c_{10}) \quad \text{or} \quad (c_9 x_2 + \cos(x_0)) \quad (138)$$

$$\psi(\mathbf{x}) = (x_2 + \cos(x_0)) \quad \text{or} \quad \sin(x_0) \quad \text{or} \quad (x_3 \sin(x_0) c_9 \cdots + \sin(x_0 + c_{10})) \quad (139)$$

$$\mathcal{E}(\mathbf{x}) = c_{11} x_2 \cos(x_5) + c_{12} x_3 + c_{13} x_6 \quad (140)$$

$$\mathcal{N}(\mathbf{x}) = c_{14} x_0 + c_{15} x_4 \quad (141)$$

$$\mathcal{D}(\mathbf{x}) = c_{16} x_3 + c_{17} x_5 + c_{18} x_6 \quad (142)$$

Setting $\mathcal{M}(\mathbf{x}) = \sin(c_9x_0 + c_{10})$ and $\psi(\mathbf{x}) = \sin(x_0)$ to then optimize the constants using CMA-ES gives the following result:

$$\hat{y} = 2934.956x_0 + 1838.41x_2 + 1900.943x_3 + 51.009x_4 - 997.463x_5 + \frac{7583.144x_0 + 11.245x_4}{-0.125x_3 - 2.662x_5 + 5094.609x_6 + 3.401} + 0.0 \quad (143)$$

While the expression looks obviously quite complex, optimizing its constants in a 10-fold cross-validation gives a mean $R^2 = 0.7197 \pm 0.0385$, indicating a strikingly low amount of overfitting. Nevertheless, $R^2 \approx 0.72$ is quite low when compared to the original $R^2 \approx 0.96$, so either the LLM was not able to correctly detect the most meaningful similarities, or the equations are too different from each other to be reconducted to a common shape.

Looking back at Table 3, PySR with default hyperparameters obtained a mean $R^2 = 0.8773$, so we deemed it useful to also examine the simpler equations obtained in that case:

$$y = (x_5(x_3 - 6.95) + 49.1) \left(\frac{x_0 \left(x_3 \cdot 106. + 112. + \frac{124.}{1.26 - x_5} \right)}{x_0 - -0.00899} + 16.4 \right) \quad (144)$$

$$y = (x_0(x_0(-6.31) - x_5 + 20.0) - -0.564) \left(-x_2x_4 + (x_3 - -3.92) \left(-x_5 + 168. + \frac{92.1}{1.27 - x_5} \right) \right) \quad (145)$$

$$y = x_0 \left(x_3 \cdot 37.5 + \frac{x_4 + \frac{\frac{x_3}{-0.0380} - 384.}{x_0 + x_5 + 0.153}}{x_5 - 1.25} \right) 55.3 - -1.17 \cdot 10^3 \quad (146)$$

$$y = x_0 \left(822. - \frac{535.}{x_5 - 1.27} \right) \left(x_3 - x_5 + \frac{11.6}{x_0 - -0.130} \right) \quad (147)$$

$$y = \frac{(x_0(4.28 \cdot 10^3 - (x_4 + x_5 \cdot 16.3)21.4) + 0.875) \left(\frac{x_3}{0.466} + 2.44 + \frac{2.35}{1.29 - x_5} \right)}{x_0 + 0.0105} \quad (148)$$

$$y = x_0(1.94 \cdot 10^3 - x_0 \cdot 642.) \left(x_3 \cdot 2.96 - x_5 - -6.40 - \frac{3.82}{x_5 - 1.27} \right) + x_4 \quad (149)$$

$$y = x_0 \left(x_3 \cdot 2.52 \cdot 10^3 + \frac{3.40 \cdot 10^4 - x_0(x_4 - -48.4)95.9}{x_2 + x_5(-1.09)x_5x_5} \right) - -1.11 \cdot 10^3 \quad (150)$$

$$y = (x_3 + 3.27) \left(x_0 \left(x_4 (x_3 - x_5) + \frac{2.10 \cdot 10^3 - \frac{1.80 \cdot 10^3}{50.5 - (x_5 + 49.2)}}{x_0 + 0.147} \right) + 2.46x_4 \right) \quad (151)$$

$$y = -x_4 + (3.11 - x_0)(x_3 - x_5 + 7.70) \left(x_0 \left(173.x_3 + 401. + \frac{346.}{1.27 - x_5} \right) - x_6 - 34.6 \right) \quad (152)$$

$$y = \left(x_0 \left(x_0 + \frac{1.16 \cdot 10^3}{x_0 + 0.0315} \right) - x_4 + 122. \right) \left(x_3 \cdot 3.41 - x_5 - 22.1 + 28.8 - \frac{4.34}{x_5 - 1.27} \right) \quad (153)$$

Answering again the prompt in Textbox E with the new files, the text produced by the LLM gives these extra insights:

1. The most striking structural similarity across almost all folds is the identification of a rational term with a denominator highly dependent on x_5 approaching a value between 1.25 and 1.3. This suggests the underlying data has a sharp peak, asymptote, or non-linear transition near this threshold;
2. Similarly, multiple folds identify a secondary instability or necessary feature transformation when x_0 is close to zero. This manifests as another rational term dividing by $(x_0 + \epsilon)$, where ϵ is a small constant;
3. The overall topology of the equations tends to favor a product-of-sums, with x_0 frequently acting as a primary global scaling factor that sits entirely outside the heaviest parenthesis blocks. Folds 2, 3, 5, and 6 all explicitly factor x_0 to the very front of the main expression, multiplying the rest of the generated terms;
4. The evolutionary search frequently groups x_3 and x_5 together in linear combinations before passing them through further operations.

Again, a generic equation was not produced at the end of the first prompt, so the second prompt was used and the answer was:

$$\hat{y} \approx \left(c_1 x_0^2 + c_2 x_0 + \frac{c_3}{x_0 + c_4} + c_5 x_4 + c_6 \right) \cdot \left(c_7 x_3 - c_8 x_5 + \frac{c_9}{c_{10} - x_5} + c_{11} \right) + c_{12} x_4 + c_{13} \quad (154)$$

Optimizing the constants with CMA-ES generates the following equation:

$$\hat{y} = 92.431x_4 + \left(25.733x_3 - 8.269x_5 + 51.408 + \frac{36.008}{1.286 - x_5} \right) \cdot \quad (155)$$

$$\cdot \left(-72.759x_0^2 + 220.591x_0 - 1.82x_4 + 131.606 - \frac{414.259}{x_0 + 274.477} \right) - 6061.062, \quad R^2 = 0.8909 \quad (156)$$

In a 10-fold cross-validation, the mean performance is $R2 = 0.8901 \pm 0.0223$, indicating almost a negligible amount of overfitting. It is also interesting to notice that this value is slightly higher than the original $R2 = 0.8773$. Due to PySR's default settings, no functions heavily employed in the first set of equations, such as $\sin$ or the exponential, were used: But the good performance of these simpler equations led us to believe that it might be possible to obtain even better results by using PySR's default settings and just add more operators. However, this did not work, with a mean $R2 = 0.8808$. Another run with `maxsize=60` and `maxiterations=1000`.

E.11 Task 361237, enhanced hyperparameters

In task 361237 (`concrete_compressive_strength`) SR obtains a remarkable mean performance $R2 = 0.8612$. Even if this good result is significantly worse than the result obtained by the two other algorithms, with XGBoost being the best at $R2 = 0.9317$, we deemed it interesting to analyze the equations obtained:

$$y = \frac{2.69x_4 \log(x_0)}{0.246 - 0.282x_3} + \frac{\log(x_5) - 0.775}{-29.0 \sin(x_7) - 13.9} - \frac{e^{\sin\left(\frac{\cos(\log(x_7(x_2-x_3x_7)(-x_4-0.877))-0.683)}{\sin(\log(x_0)-0.306)}\right)} - \sin(1.97e^{\sin(1.36x_3 - \sin(x_1x_3))})}{\cos(\sin(x_1 \sin(x_3)))} \quad (157)$$

$$y = 0.0707x_0 + \sin(x_3) - \sin(1.40x_3) - \sin(x_4) - \sin(0.268x_1x_4) - 12.2 + \frac{0.292x_0 \sin\left(\frac{1.36x_3}{-1.41x_4 + \cos(x_2 + \sin(4.33x_4))}\right)}{\quad} \quad (158)$$

$$y = \frac{2.54x_0 \left(e^{-\sin\left(\log\left(x_6 + x_7 \left(\frac{2.78x_0(\log(x_7(x_2x_5x_6^2 + 0.00209)) - \cos(1.71x_3) + \cos(\sin(x_5x_6)) + 1.33e^{-\sin((x_7+0.0589)\log(x_6) + \sin(\sin(x_6)))}{x_3} \right)} \right)} \right)}{\quad} \quad (159)$$

$$y = x_4 + \sin(x_4 - x_7) + \sin(x_5 - \sin(-x_1 + x_3)) + \sin(-(-0.868)x_2 + x_3) + \cos(x_4) + \cos(x_7) + \cos\left(\left(x_1 + \right.\right) \quad (160)$$

$$y = 0.0892x_0 - 0.158x_3 - \left(-\frac{x_1}{x_3} + \cos(\log(0.532x_0x_7)) - 4.53 \right) \log((x_7 - 0.750)(0.0296x_0 + x_1 + 0.480x_2 - 0.00532x_3)) \quad (161)$$

$$y = 0.105x_0 + 0.0187x_1 \log(x_7) + 0.00532x_2 - 0.194x_3 + x_4 - 0.00532(x_4 - \log(x_7))(-0.0690x_4(-x_3 + \cos(x_5))) \quad (162)$$

$$y = 0.0958x_0 + \left(\log \left(\frac{(x_1 + 11.6)(x_4 + 1.43)(x_1 \log(x_1 + 0.399) + x_5) \log(x_0x_2 + x_6(0.515 + \frac{\sin((x_3 - x_6 + \log(x_7)))}{x_3}))}{x_3} \right) \right) \quad (163)$$

$$y = \left(-\cos(\log(3.92x_4 + 0.136)) + \frac{2x_0 + 2x_1 + x_2}{x_3} \right) \log \left(x_7 \left(x_3 + \frac{(x_4x_7 + x_6) \log(x_2 + 0.0346)}{x_3} \right) \right) + (0.0102x_5) \quad (164)$$

$$y = -0.0277x_7 + 1.50 \log \left(x_7^5 \left(x_0 + \frac{x_1^3 x_7^2}{x_3} \right) (x_2 + 2.76) \left(\frac{x_0}{x_3} + x_1 + x_2 \right) \right) - \sin((3.77x_1 - 3.77x_6) \cos(x_5)) - \cos(x_5) \quad (165)$$

$$y = x_0 \left(0.0293 + \frac{14.7}{x_3} \right) + \frac{-25.8 + \frac{2.22x_7 \left(-\frac{0.109x_0x_1 e^{\sin(x_5 - \cos(\log(199.x_1 + 199.x_7)))} \sin(0.0547x_6)}{x_1 - 0.0741} + 0.766x_1 + \frac{0.766(0.477 - 0.576x_2)}{-0.478x_1 + 0.687x_2} \right)}{0.0741x_7 + 0.873}}{x_3} \quad (166)$$

After prompting the LLM with the content of Textbox E, the response remarks the following structural similarities:

1. The variables x_0 , x_1 , x_2 , x_3 , x_4 , and x_7 are selected and actively utilized in all ten of the provided equations. While x_5 and x_6 are extremely common, they are not strictly universal; for instance, x_5 is omitted entirely from Fold 1, and x_6 is excluded from Fold 5;
2. Every equation heavily relies on sin and cos operators to model non-linear periodic behaviors or introduce complex oscillations;
3. The genetic programming tree frequently nests these functions deeply, creating highly non-linear transformations;
4. The log function is used across almost all folds to compress features or handle exponential scaling;

5. The most striking structural similarity is how often the variable x_3 appears in the denominator;
6. The equations consistently feature standalone scalar biases added or subtracted at the root level of the equation;
7. The variable pairs (x_1, x_3) and (x_0, x_3) are frequently coupled together algebraically, interacting through addition, multiplication, or division before being passed into non-linear transformations.

No equation was produced at the end of the text, so the second prompt was used, obtaining:

$$\hat{y} \approx c_0 + \sum_{i \in \{0,1,2,4,7\}} c_i x_i + \frac{f_{\text{lin}}(x_0, x_1, x_2)}{c_{d1}x_3 + c_{d2}} + c_l \log(c_{l1}x_7 + c_{l2}) + \sum_k c_{s,k} \sin(\theta_k) + \sum_m c_{c,m} \cos(\phi_m) \quad (167)$$

Where the phase arguments θ_k and ϕ_m take either a linear or nested form:

$$\theta, \phi \in \{c_a x_j + c_b, \quad c_a x_j + c_b \sin(c_p x_q)\} \quad (168)$$

And $f_{\text{lin}}(x_0, x_1, x_2)$ is a linear function. A manual rewriting of this extremely complex expression gives:

$$\begin{aligned} \hat{y} \approx & c_0 + c_1 x_0 + c_2 x_1 + c_3 x_2 + c_4 x_4 + c_5 x_7 + \frac{c_6 x_0 + c_7 x_1 + c_8 x_2}{c_9 x_3 + c_{10}} + \\ & + c_{11} \log(c_{12} x_7 + c_{13}) + c_{14} \sin(c_{15} x_1 + c_{16} x_3) + c_{17} \cos(c_{18} x_1 + c_{19} x_3) \end{aligned} \quad (169)$$

After using CMA-ES to optimize the constants, we obtain:

$$\begin{aligned} \hat{y} = & 0.038x_0 + 0.01x_1 - 0.047x_2 - 0.017x_4 - 0.072x_7 + 13.513 \log(0.647x_7 + 1.877) + \\ & + 0.479 \sin(17.019x_1 + 11.752x_3) - 0.602 \cos(22.689x_1 + 44.699x_3) - 41.483 + \\ & + \frac{19.59x_0 + 20.475x_1 + 30.256x_2}{1.697x_3 - 32.863} \quad R2 = 0.8242 \end{aligned} \quad (170)$$

Optimizing the constants in a 10-fold cross-validation gives a mean $R2 = 0.8218 \pm 0.0286$, indicating a relatively low amount of overfitting; however, the standard deviation is also high, when compared to other tasks; and the performance of Eq. 170 is still lower than the original $R2 = 0.8612$, and not that much better than the mean result obtained by SR using default hyperparameters, $R2 = 0.8092$. It could thus be interesting to also analyze the simpler equations obtained with the original settings:

$$y = (x_0 - x_1(-0.761))0.111 + (-(-0.0411)x_3 - 11.8) \left(-0.0124x_2 + \frac{x_3}{x_7 - -17.9} - 6.95 \right) - 15.8 \quad (171)$$

$$y = \left(0.632 - \frac{0.365}{x_7} \right) \left(\frac{x_3}{(-1)0.150(x_7 + 20.6)} + x_4 + \frac{\frac{x_0}{0.637} + x_1}{11.0} + 33.1 \right) \quad (172)$$

$$y = \left(x_1 + \left(-x_0 + x_3 + (x_3 + x_4 + x_4) 42.1 \frac{1}{x_7 + 18.7} \right) (-1.47) + \frac{17.5}{x_4 - 0.123} \right) 0.0667 + 43.5 \quad (173)$$

$$y = -8.24 - -16.5 + \frac{x_0 \cdot 20.8 + x_2 x_5 \cdot 0.0162 + \left(x_1 - \frac{6.55 \cdot 10^3}{x_7 - -14.3} \right) 16.4}{x_3} - \frac{x_2}{x_0} \quad (174)$$

$$y = 19.2 - \frac{643.}{x_7 + 14.7} + \frac{x_0 \left((x_1 + x_2) \frac{1}{2.68} \cdot 0.0746 + 16.1 \right) + \left(\frac{x_1}{0.552} + x_2 \right) 3.87}{x_3} \quad (175)$$

$$y = x_0 \cdot 0.108 + x_1 \cdot 0.0811 + \frac{x_2}{17.7} + x_3 (-0.241) - -57.9 - \frac{666.}{x_7 - -15.4} \quad (176)$$

$$y = \frac{x_0}{9.36} + \frac{x_1}{12.3} + \frac{x_2}{17.6} - \left(-(-0.145) x_3 + \frac{x_3 + 1.70}{\frac{1}{3.39} (x_7 + 14.5)} \right) + 39.9 \quad (177)$$

$$y = \frac{x_2}{1.37 + \frac{547.}{x_7}} + \frac{\frac{x_0}{-0.790} - x_1}{\frac{1}{-17.9} \left(x_3 + \frac{x_5}{x_0 \cdot 2.11 \frac{1}{x_3} + x_7 - 3.23} \right)} \quad (178)$$

$$y = x_2 (-0.0409) - 9.08 + \left(x_0 + \frac{x_1}{1.25} + x_2 + 7.13 \right) 33.9 \frac{1}{x_3 - -24.8 + \frac{2.60 \cdot 10^3}{x_7 - -7.79}} \quad (179)$$

$$y = 15.9 + \frac{10.9}{\frac{x_7}{(-0.0823)(x_2 + x_2 + x_6)} - 0.254} + \frac{x_0 \cdot 18.6 + \frac{x_1 + x_2}{0.0739}}{x_3 - -0.932} \quad (180)$$

Answering the prompt in Textbox E with Eqs. 171-180, the LLM produces text highlighting the following similarities:

1. The variables x_0 , x_1 , x_3 , and x_7 are present in all 10 equations. This strongly suggests they are the primary drivers of your target variable;
2. The variable x_2 appears in 8 out of the 10 folds, missing only in folds 1 and 2;
3. Variables x_4 , x_5 , and x_6 appear very sparsely. x_4 only shows up in folds 1 and 2; x_5 in folds 3 and 7; and x_6 exclusively in fold 9. These are likely capturing fold-specific variance or noise;

4. Almost every equation places x_7 in a denominator, establishing an inverse relationship. Even more interestingly, it frequently takes the form of $\frac{\text{constant}}{x_7+k}$ or a similar variation. The offset constant k is remarkably stable across the folds, typically clustering between 14 and 20;
5. Linear Combinations of x_0 , x_1 , and x_2 : These three variables generally act as the linear foundation of the model. They are often scaled by small constants and added together before being subjected to the more complex non-linear terms;
6. While x_7 is usually isolated in its own rational term, x_3 frequently acts as a global dividing factor for larger groupings of variables.

Again, no generic equation was returned, so the second prompt was given in input to the LLM, obtaining the following:

$$\hat{y} \approx c_0 + \frac{c_1x_0 + c_2x_1 + c_3x_2}{c_4x_3 + c_5} + \frac{c_6}{x_7 + c_7} \quad (181)$$

Optimizing the constants using CMA-ES generates the following expression:

$$\hat{y} = 13.342 - \frac{739.372}{x_7 + 17.038} + \frac{627.204x_0 + 477.672x_1 + 379.5x_2}{27.052x_3 + 801.264} \quad R2 = 0.8179 \quad (182)$$

And in a 10-fold cross-validation we obtain a mean $R2 = 0.8154 \pm 0.0272$, which is surprisingly close to the previous $R2 = 0.8218 \pm 0.0286$ of the much more complex Eq. 170! The two equations share the building block $\frac{c_1x_0+c_2x_1+c_3x_2}{c_4x_3+c_5} + c_0$. In fact, this block alone, with constants re-optimized by CMA-ES, can obtain a mean $R2 = 0.4404 \pm 0.0455$.

E.12 Task 361258, enhanced hyperparameters

In task 361258 (kin8nm), SR manages to obtain a mean performance of $R2 = 0.7440 \pm 0.0435$, which is not separable from the values of RF and XGBoost: No algorithm performs significantly better or worse than the others. The equations found by SR during the 10-fold cross-validation are:

$$y = \sin \left((x_2 + 1.23) 0.197 + ((x_0 + (x_1 (-0.394) + x_4 \cdot 0.545 - x_4 (-1.13) - x_7 + x_7 - (x_1 + x_3x_4) + (x_3 + x_4 + \right. \quad (183)$$

$$y = \cos \left(\sin \left(\left(\frac{x_6 (-x_3 - 0.802)}{-1.81} - (x_2 (-1.23) + (x_3 (-(-1.04) (x_5 - x_7) - 0.375) - x_4) (-0.477) - 0.262) \right) \right) \right) \quad (184)$$

$$y = \cos \left(\sin \left(e^{\left(x_4 - (x_2 + x_2) - \left(\frac{\cos(x_0 + x_5 - (0.911 - (x_2 + x_3))(-x_2 + \sin(x_5 + \frac{x_6}{0.651} - (-0.553(x_3 + x_4 - (x_0 - (x_5 + x_5 + x_6 - \frac{x_7}{0.478})) - 0.795) + \right)} \right)} \right) \right) \right)$$

[illegible]

$$y = \sin \left(\left(-10.5x_2 + \frac{x_7}{e^{x_7}} - \left(x_0 - \left(\frac{x_4x_5(-x_6-x_6+x_7) - (-x_2+(x_5+x_6)((-x_1-0.508)(-0.471)\frac{1}{e^{x_7}} - \right)}{\right)} \right) \right) \right)$$

$$y = \cos\left(\cos\left(-0.0385\left(x_0 + \left(-0.383x_4 + \left(x_3 + ((-1.60(0.252 - x_4) - 1.53)1.47 + 2.16)\right(-(x_1 - (x_2 - 1.05\right.\right.\right.\right.$$

(188)

$$y = \cos \left(\frac{\cos \left(\left(\left(x_0 + x_2 - (x_3 + (\cos((x_2 + x_5 + x_6 - (x_2 + x_2 + x_4))(-0.426))(-(-x_7 + 0.982 + 1.86) - 1. \right. \right. \right. \right. \right.}{\hspace{10cm}} \tag{189}$$

$$y = \cos\left(\cos\left(x_2 \cdot 0.387 + \left(\left(x_0 - x_4 + \left(x_6 \left(x_4(x_2 - 1.66) \left(x_6 + \frac{x_4 + x_5 - (x_4 - x_5 + x_7)}{0.671}\right)\right) - (0.796 - (x_5\right.\right.\right.\right.\right.$$

(190)

$$y = (x_2 (-0.177) + (((0.623 - (x_2 + x_5) 0.396) 0.320 + \cos(x_1 (-1.01) + (x_0 + 0.230) x_2 (-0.240) - 0.478)) x_0 +$$

(191)

$$y = -0.0429x_0 + \left(\sin(x_2) + \sin \left(\left(x_4(0.799 - x_2) - x_5 - x_6 + \frac{-(-x_5 - x_6 + x_7 + (1.40 - x_3)(1.40 - x_3) \sin(x_2))}{(1.40 - x_3)^2} \right) \right) \right) \quad (192)$$

Answering the prompt in Textbox E, the LLM produces text highlighting the following structural similarities:

1. Variables $x_2, x_3, x_4, x_5, x_6, x_7$ appear in equations for all folds;
2. x_0 appears in 9/10 folds;
3. In almost every fold where it is isolated, the global coefficient for x_2 converges to approximately -0.155 to -0.164 ;
4. Similarly, x_0 consistently acts as a linear offset with a scaling factor hovering tightly around -0.044 ;
5. Variables x_5 and x_6 are nearly inseparable. They repeatedly appear summed together as an entry point into a trigonometric function;
6. Whenever the $x_5 + x_6$ block appears, x_3 and x_7 are typically nearby acting as phase shifts or multipliers inside the nested sin/cos expressions.

The text ends with the following generic equation:

$$\hat{y} \approx C_0 - 0.16x_2 - 0.044x_0 + g(x_3, x_4, x_5 + x_6, x_7) \quad (193)$$

where $g(\cdot)$ is a periodic function (or nested composition of cosines/sines) heavily dependent on the invariant sum $x_5 + x_6$, modulated by x_4, x_3 , and x_7 . After a human intervention, the equation now has the following shape:

$$\hat{y} \approx c_0 + c_1x_0 + c_2x_2 + \sin(x_6x_7(c_3x_3 + c_4x_4)) \quad (194)$$

But let's also take a look at the simpler equations generated by SR using default hyperparameters:

$$y = (x_4 - x_6(x_3 + x_5(x_4 + 0.504) + 0.591) - (x_2 + x_2 + (x_0 + x_3x_5)0.647))0.0744 + 0.719 \quad (195)$$

$$y = 0.394 + (x_0 + x_5 + x_6 - (2.13 - x_2) \left(x_4 - x_6 \left(\frac{x_3}{1.59} + x_5(x_4 - -0.415) \right) + 3.98 \right)) (-0.0383) \quad (196)$$

$$y = 0.732 - (x_2 + x_2 + (x_3 + 0.592)(x_5 + x_6 - -0.343) - (x_4 - (x_0 + x_5x_6)0.554))0.0709 \quad (197)$$

$$y = 0.719 - \left(x_2 - -0.402 \left(x_3x_5 - x_4 + x_6 \left(x_3 - \left(\frac{x_5 \left(x_4 + \frac{0.458}{x_6} \right) - 0.727}{-0.708} \right) \right) \right) \right) 0.155 \quad (198)$$

$$y = (x_0 + (x_2 - x_4)3.26 + (x_5 + x_6)(x_3 + (x_4 + 0.378)(x_5 + x_6 - x_7) + 0.898))(-0.0466) - -0.747 \quad (199)$$

$$y = -0.152x_2 + ((x_3 + 0.743)(x_5 + x_6 - x_7 \cdot 0.534 - 0.416) - (-0.685x_0 + x_4 \cdot 1.21))(-0.0564) + 0.733 \quad (200)$$

$$y = 0.737 - (x_2 \cdot 0.154 + (x_0 - x_4 \cdot 2.18 + (x_3 + x_5(x_4 + 0.473) + 0.691)(x_5 + x_6 - x_7))0.0513) \quad (201)$$

$$y = 0.718 + (x_0 - x_2(-3.04) + x_3(x_5 + x_6) + x_4(x_5x_6 - (1.38 - x_2)) + x_5 + x_6)(-0.0508) \quad (202)$$

$$y = (x_2(-2.33) - (-x_4 - x_4 + (x_0 + x_1)0.487 + (x_5 + x_6)(x_3 + x_4x_5 + 0.719)))0.0636 + 0.730 \quad (203)$$

$$y = 0.727 - \left(-x_2 + x_4 + \left(x_3 + \frac{x_4(x_5 + x_6 - x_7)}{1.47} - 0.637 \right) (x_5 + x_6 + 0.231)(-0.437) \right) (-0.150) \quad (204)$$

If we follow the equation that the LLM gives us:

$$\hat{y} \approx c_0 + c_1(c_2x_2 + c_3x_0 + c_4x_4 + (x_5 + x_6 + c_5x_7 + c_6)(x_3 + c_7x_4x_5 + c_8x_4 + c_9)) \quad (205)$$

Optimizing its constants with CMA-ES leads to the following equation:

$$\begin{aligned} \hat{y} = & -0.042x_0 - 0.153x_2 + 0.126x_4 - \\ & -0.061(x_3 + 1.129x_4x_5 - 0.054x_4 + 0.667)(x_5 + x_6 - 0.611x_7 + 0.246) + 0.72, \\ & R2 = 0.5648 \end{aligned} \quad (206)$$

The performance in a 10-fold cross-validation is $R2 = 0.5640 \pm 0.0234$, higher than the mean $R2 = 0.5241$ of the original SR with default hyperparameters. Taking an equation from the most complex set and optimizing the constants gives $R2 = 0.7704$ with a mean $R2 = 0.7699 \pm 0.0119$ in a 10-fold cross-validation.

$$\begin{aligned} \hat{y} = & -0.05x_0(-0.08x_2 - 0.08x_5 + \cos(1.142x_1 + 0.457x_2(x_0 + 0.779) + 0.566) + 0.466) - \\ & -0.158x_2 + 0.02x_4 + 0.021(-1.541x_1 + x_4 + 0.393x_7 + 0.138(x_2 + x_7) \cdot \\ & \cdot (-2.315x_4(0.018x_1 + x_6) + 4.536) + 0.007(-54.579x_4 - 12.551)(-3.126x_1 + 12.264x_2 - 15.621) - \\ & -1.541\cos(x_3) + \cos(x_4) + \cos(24.105x_3(0.422 - 0.91x_4)) + 55.075)\cos(1.049x_5 + 1.02x_6 + 0.018x_7) - \\ & -1.213\sin(0.015x_0 + 1.051x_5 + 1.02x_6 + 1.533) - \\ & -0.272\cos(x_2 - 0.069)\cos(x_3 + x_5 + 0.954x_6 - 0.637x_7 + 3.552) + 0.689 \end{aligned} \quad (207)$$

So, in this case we can conclude that the LLM is failing to capture the common parts of the equations which actually matter; or the equations are so different from each other that the task is more difficult than usual.

Asking the LLM again after running the SR equations through a more refined simplification process based on equality saturation graphs from the `SymbolicUtils.jl`, gives the following form:

$$\hat{y} \approx c_0 + c_1 x_0 + c_2 x_1 + c_3 x_2 + c_4 x_4 + c_5 \cdot \sin \left(c_6 + c_7 x_1 + c_8 x_2 + c_9 x_3 + c_{10} x_5 + c_{11} x_6 - c_{12} x_7 \right) + c_{13} \cdot \cos \left(c_{14} + c_{15} x_3 + c_{16} x_5 \right) \quad (208)$$

Optimizing the constants gives a mean $R2 = 0.6187 \pm 0.0190$, in a cross-validation:

$$\hat{y} = -0.043x_0 - 0.025x_1 - 0.149x_2 + 0.071x_4 - 0.197 \sin(0.06x_1 + 0.165x_2 + 0.94x_3 + 0.928x_5 + 0.921x_6 - 0.608x_7) \quad (209)$$

A manual merge of Eqs. 206 and 209 gives:

$$\begin{aligned} \hat{y} \approx & c_0 + c_1 x_0 + c_2 x_1 + c_3 x_2 + c_4 x_4 + \\ & + c_5 \sin(c_6 + c_7 x_1 + c_8 x_2 + c_9 x_3 + c_{10} x_5 + c_{11} x_6 + c_{12} x_7) + \\ & + c_{13} \cos(c_{14} + c_{15} x_3 + c_{16} x_5 + c_{17} x_6 + c_{18} x_7) + \\ & + c_{24}(c_{25} x_5 + c_{26} x_6 + c_{19} x_7 + c_{20})(c_{27} x_3 + c_{21} x_4 x_5 + c_{22} x_4 + c_{23}) \end{aligned} \quad (210)$$

Optimizing the constants with CMA-ES:

$$\begin{aligned} \hat{y} = & -0.043x_0 - 0.026x_1 - 0.154x_2 + 0.108x_4 + \\ & + 0.035(0.443x_3 + 1.556x_4 x_5 + 0.07x_4 + 0.343)(-0.811x_5 - 1.535x_6 + 0.788x_7 + 0.046) - \\ & - 0.008 \sin(0.376x_1 - 0.088x_2 + 2.919x_3 + 3.735x_5 + 1.662x_6 + 4.165x_7 - 3.351) + \\ & + 0.161 \cos(1.004x_3 + 1.031x_5 + 0.969x_6 - 0.657x_7 + 0.69) + 0.691 \\ R2 = & 0.6944 \end{aligned} \quad (211)$$

With a performance in a 10-fold cross-validation of $R2 = 0.6939 \pm 0.0163$, indicating a small amount of overfitting.

E.13 Task 361259, enhanced hyperparameters

In task 361259 (`pumadyn32nh`), SR performs significantly better than the other two algorithms, showing a mean performance of $R2 = 0.6585 \pm 0.0159$, with the second best being RF with $R2 = 0.6474 \pm 0.0176$. The equations discovered during the 10-fold cross-validation are:

$$y = \cos(x_4) x_{15} (-0.00107) \quad (212)$$

$$y = x_{15} \cos(x_4) (-0.00107) \quad (213)$$

$$y = \left(x_{15} - \left(x_2 + x_4 + \frac{0.650}{x_{16}} \right) 4.87 (0.250 - x_{11} \cdot 0.105 x_{28}) \right) \cos(x_4) (-0.00107) \quad (214)$$

$$y = \cos(x_4) (-0.000850) \frac{1}{-0.000169 - \frac{0.786}{x_{15}}} \quad (215)$$

$$y = \frac{\left(x_{15} (-0.000195) - \frac{6.57 \cdot 10^{-5}}{0.489 \frac{1}{-(-x_{19} + x_4) + x_{19} (-0.430) \frac{1}{x_{16}}}} \right) \cos(x_4)}{0.182} \quad (216)$$

$$y = \cos(x_4) x_{15} (-0.00107) \quad (217)$$

$$y = ((-x_{28} + \cos(x_{16})) x_{20} \cos(x_5) x_{15} \cdot 0.0115 + x_{15} \cdot 0.779 \cos(x_4) - \sin(x_2)) (-0.00141) \quad (218)$$

$$y = \left(-(-0.0103) x_{12} x_{30} (-x_{21} + x_{25} x_{29}) \left(x_5 + \frac{x_{31}}{x_{24}} \right) + x_{15} - x_4 \right) \cos(x_4) (-0.00108) \quad (219)$$

$$y = (x_{15} \cos(x_4) - \sin(x_2)) (-0.00108) \quad (220)$$

$$y = -0.00108 (x_{15} - x_4 - 0.222) \cos(x_4) \quad (221)$$

With a few exceptions, the equations are surprisingly simple, and even before starting the LLM-assisted analysis, it is possible to discern a few interesting patterns, such as the frequent occurrence of just x_4 and x_{15} and the use of transcendental functions, especially $\cos$. Nevertheless, using the approach illustrated in Section E, we prompt the LLM and obtain a text highlighting the following similarities:

1. Folds 0, 1, and 5 independently discovered the exact same, simplistic equation (just slightly rearranged in Fold 1): $\hat{y} = -0.00107 \cdot x_{15} \cdot \cos(x_4)$;

2. The more complex equations are essentially this baseline model with additional terms acting as additive or multiplicative adjustments. Instead of just multiplying x_{15} by $\cos(x_4)$, the longer equations substitute x_{15} with a more elaborate parenthetical group like $(x_{15} - \text{modifier})$ before multiplying by the core $\cos(x_4)$ and the constant factor;
3. x_4 and x_{15} appear in all equations. While not universal, x_2 appears in three of the more complex folds as a modifying term. Specifically, Folds 6 and 8 subtract $\sin(x_2)$ from the core equation;
4. There is a highly stable scaling factor distributed across the models.

The equation produced at the end of the text is:

$$\hat{y} \approx c_1 \cos(x_4) (x_{15} + c_2 \sin(x_2) + c_3 x_4 + c_4) \quad (222)$$

Which, however, seems too complex given the remarks above. Optimizing its constants using CMA-ES gives:

$$\hat{y} = -1.07 \cdot 10^{-3} (x_{15} - 1.116x_4 - 1.24 \sin(x_2) + 0.08) \cos(x_4), \quad R2 = 0.6622 \quad (223)$$

And in a 10-fold cross-validation, we obtain a mean performance of $R2 = 0.6620 \pm 0.0161$, indicating a relatively small amount of overfitting. Seeing that the form $\hat{y} \approx c_0 x_{15} \cos(x_4)$ is recurrent in multiple folds, optimizing its only constant c_0 gives:

$$\hat{y} = -1.077 \cdot 10^{-3} x_{15} \cos(x_4), \quad R2 = 0.6615 \quad (224)$$

With a mean performance of $R2 = 0.6612 \pm 0.0157$ over the 10-fold cross-validation, indicating again a minimal amount of overfitting.

E.14 Task 361267, default hyperparameters

In task 361267 (**brazilian_houses**), no algorithm is significantly better or worse than the others, due to a large standard deviation in performance between folds. SR still shows the highest mean $R2 = 0.6718 \pm 0.3171$ (with the second best, RF, at $R2 = 0.3828 \pm 0.3372$). Even when using enhanced hyperparameters, SR's mean performance only grows slightly to $R2 = 0.6937 \pm 0.2696$, so for this analysis we will focus on the equations found while employing default hyperparameters:

$$y = x_3 \cdot 1.29 \cdot 10^3 + x_4 \cdot 388. + x_8 + (x_0 + x_6) \left(x_1 + \frac{x_1 (x_0 - x_6)}{x_3} \right) + \frac{298.}{0.472 - x_7} \quad (225)$$

$$y = x_0 \cdot 342. + x_2 \left(x_3 (-6.23 \cdot 10^4) \frac{1}{x_1 - 38.7} - 916. + x_7 (-4.50 \cdot 10^4) \frac{1}{x_1} \right) + x_3 \cdot 1.79 \cdot 10^3 + x_8 \quad (226)$$

$$y = x_1 + x_2 \cdot 282. - x_4 (-498.) - x_7 \cdot 1.21 \cdot 10^3 - (x_0 (-390.) + (-(-1.21 \cdot 10^3) x_3 + x_8) (-1.00)) - 274. \quad (227)$$

$$y = x_3 \cdot 975. + x_8 - \left(x_5 + \frac{x_7}{0.00345} \right) + \left(\frac{x_1}{x_8 + 1.68} - x_8 (-871.) \frac{1}{0.996 (x_8 + 6.69 \cdot 10^3)} \right) 19.3 \quad (228)$$

$$y = x_3 \cdot 1.38 \cdot 10^3 + x_8 + (x_4 - x_7 - (-x_0 + x_7)) 448. \quad (229)$$

$$y = x_0 \cdot 328. + x_3 \cdot 1.28 \cdot 10^3 + x_8 + (-x_4 + x_7) (-654.) \quad (230)$$

$$y = x_0 x_4 \cdot 157. - x_3 (-1.56 \cdot 10^3) + x_8 \quad (231)$$

$$y = x_3 \cdot 963. + x_8 - \left(x_0 + x_2 - x_2 \cdot 304. \frac{1}{x_1 + 22.6} + x_4 - x_7 - x_7 + 4.96 \right) (-448.) \quad (232)$$

$$y = x_4 \left(1.04 \cdot 10^3 + \frac{x_3 (x_8 - 4.13 \cdot 10^4)}{x_1} \right) + x_8 - (-x_3 + x_7) 1.51 \cdot 10^3 - 1.53 \cdot 10^3 \quad (233)$$

$$y = -(-1.48 \cdot 10^3) x_3 + x_7 (-1.20 \cdot 10^3) + x_8 - (x_0 + x_4) (-430.) \quad (234)$$

The text produced by the LLM in response to the prompt in Textbox E highlights the following similarities:

1. The most striking similarity is that every single equation relies on the exact same linear foundation. If you strip away the more complex fractional and interaction terms, the core model is consistently:

$$y \approx C \cdot x_3 + x_8 + \text{residuals}$$

2. Frequency of appearance of the other variables: x_7 (9/10 folds), x_0 and x_4 (8/10), x_1 (6/10);

3. The models favor computing a heavy linear combination first, and then adding a smaller, non-linear component to capture variance. You rarely see deep nesting; the equations generally flatten out into sums of distinct terms;
4. When a variable recurs in a similar structural position, its coefficient tends to stay in the same order of magnitude. For example, when x_0 appears as a standalone linear term, its coefficient sits tightly in the 300s (342 in fold-1, -390 in fold-2, 328 in fold-5);
5. When division is used, it almost always takes the form of a constant divided by an expression (e.g., $\frac{298}{0.472-x_7}$ in fold-0) or a variable divided by another shifted variable (e.g., $\frac{1}{x_1--38.7}$ in fold-1). These act as localized asymptotes to correct specific data regions.

The generic equation summarizing the results is reported as:

$$\hat{y} \approx c_1 x_3 + x_8 + c_2 x_0 + c_3 x_4 - c_4 x_7 + \mathcal{F}_{\text{NL}}(\mathbf{x}) + c_0 \quad (235)$$

With $\mathcal{F}_{\text{NL}}(\mathbf{x})$ representing the heavily fold-specific interactions and rational components the symbolic regression used to fine-tune the curve. While the exact formulation changes per equation, $\mathcal{F}_{\text{NL}}(\mathbf{x})$ usually takes one of two generic forms: (i) asymptotic corrections, typically formulated as $\frac{c_k}{x_1-k}$ or $\frac{c_k}{k-x_7}$, with variable x_1 almost exclusively used in denominators to create local spikes or shifts in the data; (ii) simpler interactions, occasional direct multiplications of the secondary variables, such as $c_k(x_0 \cdot x_4)$.

In a first step, we decided to ignore the more complex interactions represented by $\mathcal{F}_{\text{NL}}(\mathbf{x})$ and just use CMA-ES to optimize the constants of:

$$\hat{y} \approx c_1 x_3 + x_8 + c_2 x_0 + c_3 x_4 + c_4 x_7 + c_0 \quad (236)$$

The result is quite surprising, because the equation with constants optimized on the whole dataset has an incredibly high R^2 :

$$\hat{y} = 356.76x_0 + 1389.623x_3 + 580.394x_4 - 1126.077x_7 + x_8 + 81.76, \quad R^2 = 0.9413 \quad (237)$$

However, in a 10-fold cross-validation the performance becomes lower at $R^2 = 0.7081 \pm 0.2373$, showing the same high variance observed for all ML algorithms. As a completely linear expression already shows signs of considerable overfitting, adding more terms is likely to be counterproductive. Instead, we perform a short ablation study, removing the terms one by one to assess their relative importance. The $c_4 x_7$ term has relatively little impact on the quality of the equation, which can then become:

$$\hat{y} = 393.222x_0 + 1399.873x_3 + 571.234x_4 + x_8 - 884.748, \quad R^2 = 0.9405 \quad (238)$$

With a performance in a 10-fold cross-validation corresponding to $R^2 = 0.7004 \pm 0.2379$.

E.15 Task 361260, enhanced hyperparameters

In task 361260 (`miami_housing`), SR performs significantly worse than the tree-based ensembles, showing a mean performance of $R2 = 0.8152 \pm 0.0364$, with the best being XGBoost at $R2 = 0.9223 \pm 0.0128$. Still, the $R2$ value is indicating a model with a good performance. The equations discovered by SR during the 10-fold cross-validation are:

$$y = 4.42x_0 \left(-\frac{63.7x_{11}x_{14}x_3}{x_8} + x_3 \right) - x_{11} + x_{14} \left(-\frac{x_7x_8 - (x_{10} + x_4 + x_9)(x_{14}x_3 \log(x_{10}) + x_3(x_0 + 35.6) - x_3)}{\left(5.39 + \frac{2.88 \cdot 10^3}{x_{14}}\right) \left(\frac{15.0x_{13}x_9}{x_3} + x_3\right) \cos(x_{14})} \right) \quad (239)$$

$$y = x_0 + x_2 - x_3 + x_5 - x_9 - (x_1x_0 \cdot 10.6 - x_{13}) + \frac{(\cos(e^{x_{14}} \cdot 10.6) + 2.72) \left(x_{14}(x_5 + 0.267) + (1.59 - x_1x_3)2 \right)}{(240)}$$

$$y = 3.46x_{10} + x_{14}(x_3(2x_0 - 6.75) + x_4 - x_6 + 0.598) + 5.04x_2 + x_3 - 2.09x_4 + \frac{17.4x_5^2(x_3 + 12.5)}{x_6x_8} + 2x_5 - 2.95 \quad (241)$$

$$y = -x_{10} + x_3 \cdot 1.87 + x_5 - x_9 + (x_8 \cdot 31.8 - 0.220) \sin \left(\frac{30.1}{x_{11} \left(2.74 + \frac{\left(\frac{x_8}{x_{11} - \frac{x_7 - 7.07}{-6.84 \cdot 10^4}} + (x_2 + 0.0783 - 0.444)(-1.35) \right)}{x_6} \right)} \right) \quad (242)$$

$$y = \frac{x_{14} \left(x_3 + x_3 + \frac{x_7 + (x_0 + 11.4)(x_2 + x_2 + x_3 + x_3) + 0.680 + \frac{x_3 \cdot 681 - (x_5 - 121 \cdot (x_{10} - 5.84 \cdot 10^3))}{\frac{1}{x_9}(x_{11}x_7 - x_5 + x_8)}}{-x_3 + x_3 + \frac{x_3}{x_0 + 13.2}} \right)}{\frac{1}{6.18(x_3 + x_3 + x_3 + x_3 + x_3)}} (x_{11} + x_3 + x_7 + x_9) - x_2 + x_3 \cdot 89.1 + x_4 + x_4 + \dots \quad (243)$$

$$y = x_3 (1.08 - x_{11}) - x_9 + \frac{(x_0 + \cos(x_{12}))(x_8 - (x_0 x_1 (-x_1 + x_{10} + x_{10} + x_2 + x_4) - x_1 (x_0 - -1.99) (x_7 + (x_0 - x_1 - x_2 - x_3 - x_4 - x_5 - x_6 - x_7 - x_8 - 0.960)))}{-x_1 + x_7 - (-x_8 + (x_{10} + x_{11} + x_{14} x_8 + x_5 - x_6 - x_7 - x_8 - 0.960))} \quad (244)$$

$$y = x_1 - x_{11} - x_{11} + x_{14} (x_3 - x_6) + x_{14} + x_3 + x_3 + \frac{x_4}{0.530} + x_5 + x_9 - \left(x_{14} - \left(\frac{x_1 x_{11}}{0.357} + 0.0251 \right) \right) - \frac{-(-159)}{c} \quad (245)$$

$$y = x_{14} (-x_3 - 98.2) (-11.1) (x_{14} x_3 + x_4 + x_5 - x_7) \frac{1}{x_7 + x_8} + \frac{x_{14} x_{14} (x_0 x_{14} + x_3)}{0.146} + x_2 - x_3 (-98.3) + \frac{x_0 (x_0 - x_1 - x_2 - x_3 - x_4 - x_5 - x_6 - x_7 - x_8 - 0.960)}{c} \quad (246)$$

$$y = x_{11} (-3.87) + x_2 + x_2 + x_2 + \left(x_1 + \left((x_1 - x_{11} - x_{14} (x_0 - x_{10} + x_{11} - x_2 - x_4 + x_9 - (x_1 + x_5) - 6.05 \cdot 10^4) - 0.545 \left(-x_{11} \left(x_3 - \frac{4.55 \cdot 10^5 \log(x_2)}{x_6} \right) + x_{14} (x_{10} + 33.8 x_3 + 33.8 e^{x_{14}} - 4.14 \cdot 10^4) + \right) \right) \right) \quad (247)$$

$$y = -x_{10} - x_{11} + 2x_2 + x_5 + \frac{0.545 \left(-x_{11} \left(x_3 - \frac{4.55 \cdot 10^5 \log(x_2)}{x_6} \right) + x_{14} (x_{10} + 33.8 x_3 + 33.8 e^{x_{14}} - 4.14 \cdot 10^4) + \right)}{c} \quad (248)$$

Answering the prompt in Textbox E, the LLM returns text highlighting the following similarities:

1. The variables $x_0, x_1, x_2, x_3, x_4, x_5, x_8, x_9, x_{10}, x_{11}$, and x_{14} appear in all 10 folds. This indicates they hold the highest predictive power for your target objective and are essential to the model's fitness;
2. The variables x_6 and x_7 are present in 9 out of 10 equations (missing only in Fold 4 and Fold 8, respectively);
3. The algorithm actively prunes x_{12} and x_{13} . x_{12} appears in 6 folds, while x_{13} is largely ignored, appearing only in Folds 0, 1, and 3;

4. Most equations feature a shallow, un-nested linear combination of variables added to the more complex fractions. For example, simple terms like $+x_2$, $+x_5$, $-x_9$, or $-x_{10}$ frequently appear at the outermost level of the expression tree (prominently in Folds 1, 2, 3, 4, 6, and 9). This suggests the target has a baseline linear relationship with these specific variables;
5. There is a recurring structural motif where x_1 governs the frequency or scaling of a periodic function, which is then used as a denominator to modulate other terms;
6. The variable x_{14} is repeatedly subjected to exponential or periodic transformations, serving as a complex scaling factor. We see $e^{x_{14}}$ explicitly in Folds 1, 3, and 9, and $\cos(x_{14})$ or $\sin(\dots x_{14} \dots)$ in Folds 0 and 8;
7. All folds exhibit deeply nested fractional structures. Variables like x_8 , x_9 , and x_{14} repeatedly appear in the denominators of these deep sub-trees. For instance, Folds 0, 2, 5, and 7 feature blocks that heavily divide by x_8 or combinations involving it. This suggests the optimization is aggressively rewarding interaction ratios or inverse relationships for these specific variables.

No generic equation is produced at the end of the text; entering the second prompt gives the following equation and accompanying text:

$$f(\mathbf{x}) = (c_1x_2 + c_2x_3 + c_3x_4 + c_4x_5 - c_5x_9 - c_6x_{10}) + (c_7x_{14}(x_3 + c_8) + c_9e^{c_{10}x_{14}}) + \frac{\mathcal{N}(x_0, x_3, x_7)}{c_{11}x_8 \cos(c_{12}x_1 + c_{13}) + c_{14}} \quad (249)$$

With the numerator $\mathcal{N}(x_0, x_3, x_7)$ highly variable across folds, acting as a “catch-all” for remaining non-linear polynomial interactions. It typically involves combinations of x_0 , x_3 , and x_7 , often interacting with logarithmic functions or nested sines depending on the specific fold.

After a human rewriting and reorganization of Eq. 249, we obtain:

$$\hat{y} \approx c_0 + c_1x_2 + c_2x_3 + c_3x_4 + c_4x_5 + c_5x_9 + c_6x_{10} + c_7x_{14}(x_3 + c_8) + c_9e^{c_{10}x_{14}} + \frac{c_{15}x_0 + c_{16}x_3 + c_{17}x_7}{c_{11}x_8 \cos(c_{12}x_1 + c_{13}) + c_{14}} \quad (250)$$

Optimizing the constants of Eq. 250 with CMA-ES gives:

$$\begin{aligned} \hat{y} = & 3.428x_{10} + 19.25x_{14}(x_3 - 276.342) + 3.698x_2 + 132.981x_3 + 2.174x_4 + \\ & + 3.332x_5 - 3.676x_9 + 0.002e^{3.707x_{14}} + 8225.43 + \\ & + \frac{-6674.663x_0 - 464.115x_3 - 3349.321x_7}{358.913x_8 \cos(1785.459x_1 - 1291.356) - 1192.136} \\ R^2 = & 0.6772 \end{aligned} \quad (251)$$

In a 10-fold cross-validation, the performance is $R^2 = 0.6746 \pm 0.0304$, indicating only a small amount of overfitting.

E.16 Task 361242, default hyperparameters

In task 361242 (superconductivity), SR performs significantly worse than the tree-based ensembles, with a mean $R^2 = 0.7113 \pm 0.0203$. , with RF showing the best performance with $R^2 = 0.9217 \pm 0.0074$. Moreover, no improvements in performance are observed even when SR employs enhanced hyperparameters. For this reason, we decided to analyze the equations from the experiment with default settings. The equations discovered during the 10-fold cross-validation are:

$$y = \frac{x_{56}x_{70}}{x_{73} \left(\frac{x_{44}x_{44}}{(x_{29}-8.35)(x_{27}+x_{28}x_{35}(-0.580)-0.510)} + x_{74} - 1.23 \right)} + 7.01 \quad (252)$$

$$y = \frac{x_{66}}{0.0859} + x_9x_{70} \cdot 0.0112 \frac{1}{x_1 + x_{43}x_{78}} \left(\frac{x_{27}}{x_{72}} + x_{77} - 12.4 \right) \quad (253)$$

$$y = x_{25} + (-x_{44}x_{45} + x_{70} - (x_{27}(-0.485) - x_9 + 58.1)) x_{26} \frac{1}{(-43.9)(x_{74} - 0.767)} (-15.2) + 6.47 \quad (254)$$

$$y = x_{16}x_{77} - x_{45} + \frac{-x_0 + \frac{x_{27} \left(\frac{x_{16}x_{26}x_{61}x_9}{49.5} + x_{70} \right)}{1.12 \cdot 10^3}}{x_{72} - 1.23} + 5.67 \quad (255)$$

$$y = x_{25} (x_{77} + 3.65) + \frac{\left(-0.103 + \frac{x_{70}}{x_{44}x_{73}} \right) \left(-x_{30} + x_{44} \cdot 35.3 \frac{1}{x_{22}} \right)}{-\frac{0.654x_{28}}{x_{44}} - 0.454} \quad (256)$$

$$y = -x_{35}x_{35} + \frac{\frac{x_{27}x_{70} \left(\frac{x_6x_9}{\frac{1}{0.0162}x_{73}} + 0.332 \right)}{x_{74}} - x_{65} (-1.89 \cdot 10^3)}{x_{23} - x_{44} (-0.928)} \quad (257)$$

$$y = x_{15} + \frac{x_{29} (x_6 (-x_7 - 20.4) (x_{70} - 54.6) (-6.76 \cdot 10^{-5}) + x_{66})}{x_{71} - 0.203} \quad (258)$$

$$y = -(-x_6 - 3.74) + \frac{\frac{x_{56} \left(\frac{x_{44}}{-2.08} + \frac{x_{70}+x_9}{x_{72}} \right)}{(-0.715)x_{73} - 1.32}}{0.498} + x_{66} + x_{77} \quad (259)$$

$$y = -x_{25} + \frac{x_{26} (x_{70} + x_9 + x_9)}{x_{72}x_{73}} - \left(-\frac{103.}{x_{46} + x_{54}} + \frac{81.0}{x_{54}} \right) - 3.01 - \frac{317.}{x_1} \quad (260)$$

$$y = x_{30}x_{70} \cdot 85.6 \frac{1}{x_{28} - -53.1} \frac{1}{x_{72}(x_{44} - -12.1)} + x_{65}(x_{77} + 1.04) 1.90 - x_{65} \cdot 4.50 \frac{1}{x_{64}} \quad (261)$$

The text that the LLM produces answering the prompt in Textbox E claims the following similarities:

1. x_{70} (The Core Driver): This is the most critical variable, appearing in all 10 equations. x_{44} : Present in 6 equations (folds 0, 2, 4, 5, 7, and 9). x_9 : Present in 6 equations (folds 1, 2, 3, 5, 7, and 8). x_{72} : Present in 5 equations (folds 1, 3, 7, 8, and 9). x_{73} : Present in 5 equations (folds 0, 4, 5, 7, and 8). x_{77} : Present in 5 equations (folds 1, 3, 4, 7, and 9). x_{27} : Present in 5 equations (folds 0, 1, 2, 3, and 5).
2. Almost every equation is built around a complex fractional term (a rational function) where a numerator is divided by a heavily nested denominator.
3. The models frequently add a constant or a simple standalone variable outside of the main fraction to shift the baseline. For example, the +7.01 at the end of fold-0, or starting with $x_{25} + \dots$ in fold-2 and fold-4.
4. When x_{70} appears, it almost exclusively dictates the behavior of the numerator. It is frequently multiplied by other variables to scale the primary rational term (e.g., $x_{56}x_{70}$ in fold-0, or x_9x_{70} in fold-1).
5. The denominators are highly non-linear, often featuring the subtraction of constants from specific variables (e.g., $x_{74} - 1.23$ in fold-0 or $x_{72} - 1.23$ in fold-3) to create asymptotic behavior.

The text ends with the following generic equation and the accompanying explanation:

$$\hat{y} \approx y = c_0 + c_1x_a + \frac{c_2x_{70}x_b + c_3}{c_4x_cx_d + c_5x_e + c_6} \quad (262)$$

Where c_0 represents the global constant offset. c_1x_a represents the standalone linear variable offset (commonly acting like x_{25} or x_{15} in your folds). The numerator is dominated by x_{70} multiplied by a scaling variable x_b (acting like x_9 or x_{56}) and a coefficient c_2 . The denominator captures the non-linear interaction of variables like x_{44} , x_{72} , or x_{73} (represented by generic variables x_c, x_d, x_e), scaled and shifted by constants c_4, c_5 , and c_6 .

After a human intervention, Eq. 262 is rewritten as:

$$\hat{y} \approx c_0 + c_1x_{15} + c_2x_{25} + \frac{c_3x_{70}x_9 + c_4x_{70}x_{56} + c_5}{c_6c_{44}x_{72} + c_7x_{73} + c_8} \quad (263)$$

Optimizing the constants of Eq. 263 with CMA-ES results in:

$$\hat{y} = 78.469x_{15} - 78.356x_{25} + 3.467 + \frac{60.046x_{56}x_{70} + 1.456x_{70}x_9 - 1794.169}{179.632x_{72} + 81.658x_{73} - 201.841} \quad R^2 = 0.6638 \quad (264)$$

In a 10-fold cross-validation, the mean performance is $R^2 = 0.6633 \pm 0.0153$. While this performance is satisfying, it is lower than the original SR's mean

$R^2 = 0.7113 \pm 0.0203$. In an attempt to obtain an equation which includes more terms, we prompt the LLM again:

Additional LLM prompt used for this case study

Can you reformulate the equation including all terms that appear most often with the actual variables, for example: instead of c_1x_a , $c_1x_{25} + c_2x_{25}$ and so on? It's ok to have a more complex equation, I just want to capture as many recurrent terms as possible.

The text in response includes the following equation:

$$\hat{y} \approx c_0 + c_1x_{25} + c_2x_{77} + c_3x_{25}x_{77} + c_4x_{66} + \frac{x_{70}(c_5x_9 + c_6x_{27} + c_7) + c_8x_9 + c_9x_{27}}{(c_{10}x_{72} + c_{11})(c_{12}x_{73} + c_{13}) + c_{14}x_{44} + c_{15}} \quad (265)$$

Optimizing the constants of Eq. 265 with CMA-ES gives:

$$\begin{aligned} \hat{y} = & 4.94x_{25}x_{77} - 21.151x_{25} + 18.978x_{66} - 3.351x_{77} + \\ & + \frac{1010.188x_{27} + x_{70}(19.412x_{27} + 92.837x_9 - 2207.771) - 4663.478x_9}{54.219x_{44} + (123.637x_{72} - 216.296)(59.175x_{73} - 42.878) + 5527.946} + 11.917 \\ R^2 = & 0.7223 \end{aligned} \quad (266)$$

In a 10-fold cross-validation, the mean performance is $R^2 = 0.7217 \pm 0.0143$, indicating only a small amount of overfitting.

E.17 361247, default hyperparameters

On task 361247, SR performs significantly worse than the other two approaches. The performance with enhanced hyperparameters is actually even worse than with default hyperparameters, highlighting possible issues. The equations discovered by SR for the different folds are:

$$y = x_{11} \cdot 10.1 - 8.81 - \frac{0.00999}{7.50 \cdot 10^3 \frac{1}{x_8} \frac{1}{\frac{x_{11}x_4}{x_8(x_9 - 0.743)0.220} + x_8}} \quad (267)$$

$$y = x_8(-0.00297) + \left(x_{10} \cdot 2.89 \cdot 10^{-5} - \frac{0.00479}{x_{10}} \right) \left(x_7 \frac{1}{-0.00300x_{10}(x_5 - 0.113) + x_{10}} (-0.00472) + x_8 \right) + 2.9 \quad (268)$$

$$y = \frac{\left(x_{11} - \left((x_0 + x_9 + x_9) x_9 \frac{1}{x_{12} + x_2} x_0 \cdot 1.52 + \frac{1.25}{x_{11}} \right) \right) (x_{13} - -4.14) + 2.01}{1.13} \quad (269)$$

$$y = 0.356 - \frac{(x_9 + 0.224) \left(\left(x_{11} - \frac{0.142}{x_9} \right) x_{11} x_{11} (-0.928) + x_8 \cdot 0.000671 + 0.276 \right)}{0.417 x_9} \quad (270)$$

$$y = - \frac{-(x_8 (-0.000189) - (-x_{11} - 3.08) - 3.87) + \frac{0.788}{x_2}}{0.114} - 0.0331 + \frac{x_8 \frac{1}{x_9} (-0.000389) + 0.0807}{x_9} \quad (271)$$

$$y = -0.437 + \left(\frac{x_{10}}{0.375} + \frac{x_2}{1.07 x_7} + 233. \right) 3.09 \frac{1}{x_8} \quad (272)$$

$$y = x_{11} \left(\frac{x_{11} (x_8 + (x_0 + 44.8) 21.1)}{x_8} - 0.290 \right) - 1.46 + \frac{2.12}{(-1.91) x_{10}} + \frac{x_{13}}{x_1} \quad (273)$$

$$y = x_{10} \cdot 0.520 \left(0.112 - x_3 \cdot 8.33 \cdot 10^{-6} \frac{1}{x_9} x_9 \right) + (x_8 \cdot 0.000363 - 0.308) (-6.65) + \left(\frac{x_0}{0.134} - x_3 \right) 4.79 \cdot 10^{-5} \quad (274)$$

$$y = \frac{x_{11}}{0.431} - x_8 (-0.000636) \frac{1}{x_{11} (0.713 - x_{11})} + \frac{0.101}{\frac{1}{-2.95} x_9} \quad (275)$$

$$y = \frac{-(-0.0463) x_{10} - x_8 \left(x_4 (-5.79 \cdot 10^{-8}) \left(\frac{x_3}{-1.80} - 0.0463 \right) \frac{1}{x_3 (x_9 - 0.587)} \frac{1}{x_9} - -0.00408 \right) + 3.72}{1.62} \quad (276)$$

The text produced by the LLM highlights the following similarities:

1.

$$\hat{y} \approx c_0 + c_1 x_{11} + c_2 x_{10} + \frac{c_3 x_{11} + c_4}{x_8 (c_5 x_9 + c_6) + c_7} \quad (277)$$

However, optimizing the constants of Eq. 277 as it is gives unsatisfying results with low R^2 . After expert intervention, the new equation looks like:

$$\hat{y} \approx c_0 + c_1 x_8 + c_2 x_{10} + c_3 x_{11} + \frac{c_3 x_{11} + c_4}{x_8 (c_5 x_9 + c_6) + c_7} \quad (278)$$

Optimizing the constants of Eq. 278 gives:

$$\hat{y} = 0.028 x_{10} + 1.21 x_{11} - 0.003 x_8 + \frac{3304.762 x_{11} - 3446.023}{x_8 (3.953 x_9 - 9.853) + 3678.592} + 1.133, \quad R^2 = 0.8022 \quad (279)$$

And in a cross-validation we have a mean $R^2 = 0.8020 \pm 0.0035$.

E.18 Task 361251, enhanced hyperparameters

On this task, SR is significantly worse than the other two approaches, but it gains considerably when passing from default to enhanced hyperparameters. The equations identified by SR with enhanced hyperparameters are:

$$\begin{array}{cccccccccccccccccccc}
 y = \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & \sin & 0.0279 \log & (x_{10} \log (x_2 + 0.675) + 0.2
 \end{array}$$

(280)

$$y = \sin\left(\frac{x_{11}}{x_0 x_{11} + 1.71} + 1\right) e^{(x_2 - 0.762)0.0871 + e^{-0.839 \cdot x_2}}$$

[illegible]

$$y = 0.483 \sin \left(\sin \left(0.0167x_8 (x_8 + \log(x_3 - x_7)) \log(x_0) - 0.0681 \left(-0.896x_{11} + 0.00582(-x_2 + 2.15x_3x_8 + x_3 \right. \right. \right. \\ \left. \left. \left. (283) \right) \right) \right)$$

$$y = 0.255 \sin \left(\sin \left(\sin \left(\sin \left(\sin \left(\sin \left(\sin \left(0.0328x_{10}x_2 + 0.0146x_{11}x_3 + 0.0146x_9 \left(x_9(x_1 - 0.778) - \sin(x_1) \right) \right) \right) \right) \right) \right) \right) \right) \quad (284)$$

$$y = -\sin \left(0.169 \sin \left(\sin \left(\sin \left(\sin \left(\sin \left(\sin \left(-0.0175x_0 x_{10} x_2 x_8 + (0.0330x_{11} (0.000117x_0 x_2 x_8 + \right.\right)\right.\right.\right.\right.\right.\right.\right.\right.\right.$$

(285)

(286)

(287)

(288)

(289)

The text generated by the LLM claims the following common structural elements:

1. The variables $x_0, x_1, x_2, x_3, x_8, x_9, x_{10}$, and x_{11} are present in every single equation. This indicates that these 8 features hold the primary predictive power for your target variable y

2. The most glaring visual similarity is the heavy use of repeated, nested sine functions, i.e., $\sin(\sin(\sin(\dots)))$. This is a frequent artifact in symbolic regression when the algorithm tries to bound an output between $[-1, 1]$ or map a highly periodic or complex non-linear boundary layer
3. Every single equation constructs its inner logic using the exact same non-linear mathematical operators: natural logarithms ($\log$), natural exponentials (e^f), sines ($\sin$), and cosines ($\cos$)
4. Across nearly all expressions, x_0 repeatedly appears in the denominators of major fractional blocks. It acts as a universal scaling or normalization factor for the rest of the interacting inputs
5. You will notice specific feature pairings that constantly cling to each other across folds—for instance, x_{11} and x_2 are frequently combined within exponentials or logs, and x_1 often scales alongside x_9 .

The equation found by the LLM is:

$$\hat{y} \approx c_0 \sin \left(c_1 \sin \left(\dots \sin \left(\frac{c_2 \cdot e^{c_3 x_{11}} \cdot \log(c_4 x_2 + c_5)}{c_6 x_0 + c_7} + c_8 x_1 x_9 + c_9 x_{10} x_3 + c_{10} \right) \dots \right) \right) \quad (290)$$

With a nested set of sin functions of arbitrary length. After expert intervention, we settle on the following structure:

$$\hat{y} \approx c_0 \sin \left(c_1 \sin \left(\frac{c_2 \cdot e^{c_3 x_{11}} \cdot \log(c_4 x_2 + c_5)}{c_6 x_0 + c_7} + c_8 x_1 x_9 + c_9 x_{10} x_3 + c_{10} x_8 + c_{11} \right) \right) \quad (291)$$

Optimizing its constant with CMA-ES gives:

$$-0.0997 \sin \left(1.124 \sin \left(0.0750 x_1 x_9 + 0.06619 x_{10} x_3 + 0.3753 x_8 + 3.2948 + \frac{2.3662 e^{-0.6165 x_{11}} \log(1.0000 - 5.65)}{-1.9512 \cdot 10^{-8} x_0 - 2.0206} \right) \right) \quad (292)$$

With a performance in a cross-validation of $R^2 = 0.6499 \pm 0.0255$. While not bad, this is far from the $R^2 = 0.8148$ of the original SR's performance.

E.19 Task 361252, default hyperparameters

On task 361252 (`video_transcoding`), SR performs significantly worse than the other two algorithms. More importantly, its performance with enhanced hyperparameters is *worse* than with default hyperparameters, likely indicating issues with an incomplete exploration of the search space. The equations found by SR are:

$$y = \frac{x_{16} (x_{13} x_{13} - 0.299) 0.00328 - 0.442}{\frac{x_{13} x_{16}}{x_{15} (x_{14} \cdot 0.00328 + x_3)} + x_{13} - 0.963} - 0.211 \quad (293)$$

$$y = \frac{x_{16} \cdot 0.00297 x_{13} x_{13} - 0.582}{x_{13} - 0.963 + x_{16} \frac{1}{x_{14}} \cdot 11.7 + \frac{x_3 \left(-x_8 + \frac{x_9}{x_{15}} \right)}{x_{11}}} \quad (294)$$

$$y = x_{16} \left(-x_{13} + \frac{0.715}{x_{13} - 1.17} \right) \left(x_{15} - \frac{x_{16}}{\frac{1}{x_3} (x_{14} - -0.0512)} - x_5 (-0.448) \right) (-0.000197) - 0.0565 \quad (295)$$

$$y = \left(x_{13} (x_{17} - x_5) - \left(x_{17} (-0.163) \frac{1}{x_{13} - 2.96} - -133. \right) \right) (x_{14} \cdot 1.79 \cdot 10^{-7} - -0.652) 0.00749 \frac{1}{x_{13} - 0.881 - 0} \quad (296)$$

$$y = x_{13} (x_{13} - 0.574) - \frac{x_{16}}{x_{13} x_{15} \cdot 0.508 - 116. - \frac{1.97}{0.977 - \left(x_{13} + \frac{x_{16}}{x_{14}} \right)}} - 6.23 \quad (297)$$

$$y = \frac{x_{13} x_{13} x_{13} (x_{17} \cdot 0.00237 - 0.449)}{x_{13} - 0.976 + x_{17} \cdot 7.92 \frac{1}{x_{14}}} + x_3 \cdot 0.00790 - 1.81 \quad (298)$$

$$y = x_{16} (-x_{15} - 0.813) \left(x_{13} (-0.000191) + \frac{8.75 \cdot 10^{-5} - \frac{3.85}{x_{14}}}{x_{13} - 1.06} \right) \quad (299)$$

$$y = \left(-\frac{x_{11}}{x_{12} (x_{13} - 1.21)} + x_{13} \right) 0.792 \left(\frac{x_{16}}{208. - \left(\frac{x_{15}}{0.350} + x_3 \cdot 0.0437 + \frac{x_{12}}{x_{12}} \right)} - 0.995 \right) \quad (300)$$

$$y = x_{16} (-3.32 \cdot 10^{-6}) \frac{1}{x_{13} - 3.10} \left(\frac{x_{17} - \frac{x_{16} + x_{17}}{x_{15}}}{x_{13} - 0.973 + x_{16} \cdot 5.83 \frac{1}{x_{14}}} - 0.140 \right) + 3.56 \quad (301)$$

$$y = x_{17} (x_{14} \cdot 3.96 \cdot 10^{-9} + 0.0290) + \frac{-0.316 + \frac{x_{14} (-8.81 \cdot 10^{-7}) + x_{17} (-0.0130) + 2.66}{x_{13} - 1.67}}{x_{13} - 0.276} - 4.83 \quad (302)$$

The equation found by the LLM is:

$$\hat{y} \approx c_0 + \frac{c_1 x_{16} x_{13}^{c_2} - c_3}{x_{13} - c_4 + c_5 \frac{x_{16}}{x_{14}}} \quad (303)$$

With a constant c_2 used to determine the power of a term $x_1 x_{16} x_{13}^{c_2}$.

Optimizing its constants with CMA-ES gives:

$$\hat{y} = \frac{3.0207 \cdot 10^{-4} x_{13}^{4.1056} x_{16} - 0.0675}{x_{13} - 0.9947 + \frac{1.2066 x_{16}}{x_{14}}} + 1.4197, \quad R2 = 0.7871 \quad (304)$$

And in a cross-validation $R2 = 0.7872 \pm 0.0113$. However, having a floating point power is not something which was possible using PySR's settings; so, we empirically set $c_2 = 3$ and rerun constant optimization with CMA-ES:

$$\hat{y} = \frac{9.825 \cdot 10^{-4} x_{13}^3 x_{16} - 0.2125}{x_{13} - 0.9828 + \frac{3.8670 x_{16}}{x_{14}}} + 0.9912, \quad R2 = 0.7823 \quad (305)$$

With a mean performance in a cross-validation of $R2 = 0.7823 \pm 0.0112$, so we accept the negligible decrease in performance to obtain a simpler expression.

E.20 Task 361253, enhanced hyperparameters

In task 361253 (`wave_energy`) with enhanced hyperparameters, no algorithm performs significantly better or worse than the others. The equations discovered by SR durin the 10-fold cross-validation are:

$$y = -x_{27} + x_{31} + x_{32} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - (306)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} + \sin(x_2 - (307)$$

$$y = x_{15} + x_{28} + x_{32} + x_{33} + x_{34} + x_{35} + x_{36} (-0.00575) + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{45} (308)$$

$$y = x_{12} + x_{13} + x_{23} + x_{34} + x_{38} + x_{45} + x_{47} - (x_{16} - x_{19}) - (x_{21} - x_7) - (-x_{32} + (x_{28} - (x_{19} - -648.)) 170.) - (309)$$

$$y = -x_1 + x_{32} + x_{32} - x_{33} + x_{33} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} (310)$$

$$y = \left(x_{14} + x_{21} + x_{32} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{40} + x_{44} + x_{46} + \frac{x_{47}}{1.98} + x_{47} + \left(x_{21} + x_{21} + x_{32} + x_{32} \right) \right) (311)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} \quad (312)$$

$$y = x_0 + x_{15} + x_{16} + x_{16} + x_2 + x_{20} - x_{26} + x_{27} - x_{29} - x_{30} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + \dots \quad (313)$$

$$y = x_{21} + x_{21} + x_{21} + x_{21} + x_{23} + x_{23} - x_{25} + x_{26} + x_3 + x_{40} + x_{43} - (x_{19} + x_{41})(-0.277) + (x_3 + x_3)50.3 + \dots \quad (314)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - (-x_{40} - 0.140) \quad (315)$$

The equations are quite surprising, as they are mostly linear compositions of the features. The equation proposed by the LLM-produced text is:

$$\hat{y} \approx c_0 + c_1 x_{32} + c_2 x_{33} + c_3 x_{34} + c_4 x_{35} + c_5 x_{36} + c_6 x_{37} + c_7 x_{38} + c_8 x_{39} + c_9 x_{40} + c_{10} x_{41} + c_{11} x_{42} + \dots + c_{12} x_{43} + c_{13} x_{44} + c_{14} x_{45} + c_{15} x_{46} + c_{16} x_{47} \quad (316)$$

which basically corresponds to a linear regression.

Optimizing its constants with CMA-ES gives:

$$\hat{y} = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + \dots + x_{44} + x_{45} + x_{46} + x_{47} - 105.301, \quad R^2 = 1.0 \quad (317)$$

It's interesting to note that the coefficients $c_1, \dots, c_{11}$ are not all exactly set at 1.0, but can have values like $c_1 = 0.99995864816671$ or $c_2 = 1.00000305084385$. Optimizing the constants in a cross-validation gives $R^2 = 1.0000 \pm 0.0000$, which is of course an approximation, but still indicates an excellent fitting.

E.21 Task 361266, enhanced hyperparameters

On task 361266 (`kings_county`) SR performs significantly worse than the other algorithms. The equations discovered by SR during the cross-validation are:

$$y = x_1 + x_{14} \left(1.48x_{12} + 1.48x_{16} + 1.48x_2 + 24.6e^{x_7} + 20.2 + \frac{26.5 \left(-x_{19} \log(x_2) + \frac{x_2(x_6 - 2.77)}{x_8} \right)}{x_8} \right) + 4.18x_9 - 4 \quad (318)$$

$$y = \left(x_8 - \left(x_{18} - \left(x_5 - \cos \left(x_{14} - \left(x_8 + \frac{1.99}{x_{16}(-x_{18} - 0.678 + 1.40)} \right) \right) \right) \right) - \left(x_5 - \cos \left(x_{14} - \sin \left(\frac{1.99}{x_{16}(0.725 + 1.40)} \right) \right) \right) \right) \quad (319)$$

$$y = 0.725x_{12}x_7(1.59x_1x_8 - 1.59) - x_2x_5\left(-x_8 - 12.2 - \frac{1.97}{x_8 \sin(x_{14})}\right) \log(x_3) - x_8 + \left(-x_{15} + \left(\frac{x_2}{x_3} + x_8\right)\right) \quad (6.7)$$

(320)

$$y = (x_5 + 1.15) \left(x_8 (-x_{12} (-x_0 + \cos(x_{14})) + x_{12} + x_{16}) + \left(\frac{x_{10}}{(-0.287)^{1.11}} + x_{13}x_{13} + x_{13} + x_{16} + x_{16} + x_{18} + \right. \right. \\ \left. \left. (321) \right) \right)$$

$$y_{x_8} \left(272. - \frac{1.27 \cdot 10^4}{x_{14}} \right) \left(-(-2.28 \cdot 10^4) \left(\frac{x_{14}}{x_2} + x_5 \right) + \left(x_{12} + x_2 \left(\sin \left(-\frac{2.30 \cdot 10^4}{x_{14}} \right) + 2.01 \right) + x_6 + 2.28 \right. \right. \\ \left. \left. (322) \right) \right)$$

$$y = x_{12} + x_4 (x_0 + x_5 (-0.560)) \left(\frac{x_{12} x_6 x_8}{x_5 + \frac{x_{13} - 1.02}{2.31}} + x_{12} \right) + x_6 x_8 (x_{16} + x_{16} - 1.85) - \left(-x_7 + \left(x_{12} (-4.79) + \right. \right. \\ \left. \left. (323) \right) \right)$$

$$y = -x_{14} - 239.x_2 - x_8(x_2(-44.9) + x_7) + x_8(x_{14} + e^{0.520} - 10.7)15.6(e^{x_6} - 1.16) + (x_{14} + x_8(-22.0))((324)$$

$$y = -x_2 + \frac{x_1 \left(x_0 (-x_{19} + x_9 - 1.32 \cdot 10^3) + \frac{x_{16} x_7 \cos(\frac{x_{14} (-19.8)}{0.319}) + x_7 - 1.98}{0.633} \right) - x_{12} - x_{20} + (x_{14} \cdot 17.8 - 834.) \left(-x_0 - x_{11} + \right)}{(325)}$$

$$y = -1.95x_{11} - 3.95x_9 + 27.5(-x_1 + x_{11} + x_2)e^{x_5 - \sin(6.76e^{-\sin(x_{14})})} + (x_7 + x_8 + 14.7)\left(2x_{11} + x_{18} + x_2(x_6 - \right. \\ \left. (326)$$

$$y = x_{10} + \frac{x_{12}}{0.355} - x_9 + \left(x_{12} + x_8 \left(x_5 + \frac{x_6}{x_8 + 1.05} + 1.62 \right) \right) \left(\frac{(-1) 1.33 x_2 \left(-0.0397 + \frac{1.62}{-x_2 + x_{20}} \right) \left(x_{14} \left(\cos \left(\frac{1}{x_{14} + 1.05} \right) \right) \right)}{(327)} \right)$$

The equation proposed in the text produced by the LLM is:

$$\hat{y} \approx c_0 + c_1 x_{12} + c_2 x_2 + c_3 x_6 x_8 + c_4 e^{c_5 x_6 + c_6 x_7} + c_7 \sin(c_8 x_{14} + c_9) + \quad (328)$$

$$+ c_{10} \log(c_{11} x_2) + \frac{c_{12} x_1 + c_{13}}{c_{14} x_8 + x_{13} - x_{14}} \quad (329)$$

Optimizing its constants with CMA-ES does not produce any meaningful result, as the optimization seems unable to find good solutions, probably due to the presence of log. After a human intervention, the equation now looks like:

$$\hat{y} \approx c_0 + c_1 x_1 2 + c_2 x_1 1 + (x_2 + c_3 x_5 + c_4 x_{14} + c_5)(c_6 x_6 + c_7 x_8 + c_9) + c_{10} \sin(c_{11} x_{14} + c_{12}) \quad (330)$$

And optimizing its constants with CMA-ES gives:

$$\hat{y} = 31.867x_{11} + 74.53x_{12} + (20.352x_6 + 44.071x_8 - 226.401)(42.249x_{14} + x_2 + 2397.756x_5 - 2121.226) + 3828.56 \quad (331)$$

With a 10-fold cross-validation performance of $R^2 = 0.6434 \pm 0.0431$.

Still, the R^2 value is lower than the original. Restarting from Eq. 322 found for fold 4:

$$\hat{y} = x_8 \left(272. - \frac{1.27 \cdot 10^4}{x_{14}} \right) \left(-(-2.28 \cdot 10^4) \left(\frac{x_{14}}{x_2} + x_5 \right) + \left(x_{12} + x_2 \left(\sin \left(-\frac{2.30 \cdot 10^4}{x_{14}} \right) + 2.01 \right) + x_6 + 2.28 \right) \right) \quad (332)$$

$$\hat{y} \approx x_8 \left(c_0 - \frac{c_1}{x_{14}} \right) \left(c_2 \left(\frac{x_{14}}{x_2} + x_5 \right) + x_{12} + c_3 x_2 \left(\sin \left(\frac{c_4}{x_{14}} \right) + x_6 + c_5 \right) \right) \quad (333)$$

Asking the LLM to simplify the equation gives:

$$\hat{y} \approx x_8 \left(1 - \frac{c_0}{x_{14}} \right) \left(c_1 \frac{x_{14}}{x_2} + c_2 x_5 + c_3 x_{12} + c_4 x_6 + x_2 \left(c_5 \sin \left(\frac{c_6}{x_{14}} \right) + c_7 \right) + c_8 \right) \quad (334)$$

However, all optimizations lead to inferior results.

E.22 Task 361268, enhanced hyperparameters

On task 361268 (`fps_benchmark`), SR performs significantly worse than the other two algorithms. The equations found by SR during the cross-validation are:

$$y = x_{29} + \left(\left(x_8 - \frac{\sin(x_{30} + x_{36} - \sin(x_{36} - (x_{30} - \sin(x_{30} + x_{38}) + 0.0410) - 0.811))}{0.0498} - 0.855 \right) 1.19 - -0.859 \right) \quad (335)$$

$$y = 4.09x_1 + x_{10} + 3.54x_{20} + 2x_{24} - x_{25} + x_{29} + 2x_{30} - 3.54x_{35} + x_{38}(-1.61x_{38} - 7.46) + \log(x_6) \cos(0.0961x_{38}) \quad (336)$$

$$y = 0.846x_{18} + 20.2x_{25} + 1.95x_{30} + 0.343x_{32} - 0.846x_{36} - 6.73x_{38} + 0.585x_9 (x_{14} - 1.80) + 6.73x_9 + 6.73 \log(x_{14})$$

(337)

$$y = 0.765x_{11} + 2x_{20} + 0.765x_{29} - 15.7x_{38} + 41.6e^{\sin(\sin(-2x_{36} + \log(x_{36} + 0.0481) + 1.25))} - 5.29 + \frac{7.67e^{-x_{36}}}{x_{36} + 0.0526} + \frac{0.7}{x_{29}}$$

(338)

$$y = -0.539x_{14} + x_{19} \left(-2.13x_{35} - 2.13x_{36} \sin(x_{36}) + \cos(x_{26}) + 15.8 + \frac{12.1}{x_{38}} \right) + x_{29} - (14.7 - \log(x_{38})) \sin(x_{38})$$

(339)

$$y = 1.74x_1 - 1.74x_{18} + 7.74x_{24} + x_{29} + 1.74x_3 - 1.74e^{x_{35}} - 1.74e^{x_{38}} + 1.74e^{\cos(x_{36} + 1.87)} \log(0.580x_{34}) - 44.4e^{\cos(x_{36})}$$

(340)

$$y = x_{11} - x_{19} + x_{29} \cos(\cos(x_{38})) + 2.52x_{30} - 3x_{38} - 2.52x_9 + 5.97(2.02 - 33.0 \sin(x_{36} - 0.727)) e^{-x_{36}} + (x_{29} - 1.74x_{38}) \cos(x_{38})$$

(341)

$$y = x_1 \left(0.930 + \frac{x_{30} + 1.06}{x_{13}} \right) + 0.215x_{11} + 2x_{24} - 0.430x_{28}x_{38} + 0.859x_{29} + 0.430x_3 - 0.215x_{36}x_{38} - 0.215x_{38}^2$$

(342)

$$y = 1.76x_1 + x_{14} + x_{18} + x_{19} + x_{20}^2 + 1.76x_{20} + x_{24} + 1.76x_3 - 0.766x_{36} - 2x_{38} - 1.76x_8 \cos(2.09 \sin(2.67 \cos(x_{36})))$$

(343)

$$y = \frac{x_1}{x_{36} - 12.8 - -0.0471} + \frac{x_1}{x_{36} - -0.0471} + x_{14} - x_{18} + x_{20}x_{38} + x_{20} - x_{28} - x_{29}(x_{19} - 0.964) + x_3 + x_{38} + \frac{1.74x_{38}}{x_{36} - 12.8 - -0.0471}$$

(344)

The equation proposed by the LLM is:

$$\hat{y} \approx c_0 + \sum_{i=1}^k c_i x_i + c_{k+1} \cos(c_{k+2} \sin(c_{k+3} x_{36} + c_{k+4})) + \frac{c_{k+5}}{c_{k+6} - x_{36}} + \frac{c_{k+7}}{x_{38}} \quad (345)$$

With the most represented variables being $x_1, x_{29}, x_{30}, x_{20}$. After several iterations of human intervention, the equation now looks like:

$$\hat{y} \approx c_0 + c_1 x_1 + c_2 x_{20} + c_3 x_{29} + c_4 x_{30} + \frac{c_5}{x_{36} - c_6} + \frac{c_7}{x_{36} - c_8} + c_9 x_{38}^2 + c_{15} x_{38} x_{30} + \frac{c_{10}}{x_{38}} + c_{11} \cos(c_{12} \sin(c_{13} x_{36} + c_{14})) \quad (346)$$

Optimizing its constants with CMA-ES gives:

$$\hat{y} = 3.431x_1 + 2.782x_{20} + 0.873x_{29} + 0.333x_{30}x_{38} + 1.946x_{30} - 35.873x_{38}^2 + 39.939 \cos(25.355 \sin(6.623x_{36} - 3.141)) \quad (347)$$

And a performance in a 10-fold cross-validation of $R^2 = 0.7066 \pm 0.0099$.